EU POLICY SUPPORT TECHNICAL ASSISTANCE TO THE GOVERNMENT OF GILGIT-BALTISTAN

Policy, Institutional, Technical, and Operational Assessment

Strengthening Community-Managed Irrigation Systems for Climate Resilience and Sustainable Development

Prepared for:
Government of Gilgit-Baltistan

Funded by the European Union

Prepared by:
NTU International/Himat Consulting, Technical Assistance Team

Draft version: 03
Dated: August 28, 2026

Disclaimer: T*he views and opinions expressed are those of the author(s) and do not necessarily reflect those of the European Union or the NTU International. Neither the European Union nor the granting authority can be held responsible for them*

Document Preparation, Review and Endorsement

Key project/document information	
Project Name	SEA-2023-24956: Policy support to the Government of Gilgit-Baltistan
Country	Pakistan – Provincial Government of Gilgit Baltistan
Date	Aug 28, 2026
Version	No. 03
File name	Technical Report on Irrigation_SEA-2023-24956_v03 (Aug 2026)

Document Preparation and Review (Project Team)			
Version	**Date**	**Prepared by**	**Reviewed by**
No. 01	Mar 18, 2026	Sajid Ali Khan, Irrigation Expert	Ashab Baig, Natural Resource Management Expert & *Project Team Leader*
No. 02	May 05, 2026	Sajid Ali Khan, Irrigation Expert	*Departmental official technical review;* Ashab Baig, *Project Team Leader*
No. 03	Aug, 28, 2026	Sajid Ali Khan, Irrigation Expert	 Ashab Baig, *Project Team Leader*

Official Review and Endorsement (Departmental Team)			
Sr#.	**Name**	**Designation**	**Department/Institution**
1	Mr. Sibtain Ahmed	Secretary, Water Management and Irrigation Department	Department of Water Management and Irrigation (DoWM&I), Government of Gilgit-Baltistan
2.	Mr. Jahangir Khan	Director Water Management and Irrigation Department	DoWM&I
3.	Mr. Adil Hussain	Water Management Specialist	DoWM&I
4.	Mr. Atti Ullah	Agriculture Engineer	DoWM&I
5.	Mr. Iftikhar Ali	Agriculture Engineer & Focal person of DoWM&I	DoWM&I

TABLE OF CONTENTS

List of Tables 5
List of Figures 5
Executive Summary 6
List of Acronyms 8
1. Introduction 9
1.1 The Agroecological Context 9
1.2 Background of the TA Assignment 10
1.3 Objectives of the Irrigation Sector Study 11
1.4 Methodology and Approach 11
1.5 Stakeholder Analysis 12
1.6 Limitations and Scope 13
2. Policy and Institutional Framework 15
2.1 National Policies 16
2.2 Provincial Policies 17
2.3 Customary Law and Indigenous Institutions 18
2.4 Institutional Landscape: Roles and Mandates 20
2.5 Coordination Challenges and Opportunities 22
3. Infrastructure Quality and Design Review 26
3.1 Typology of Irrigation Systems 26
3.2 Technical Standards, Failure Patterns, and Vulnerabilities 28
3.3 Climate Resilience of Existing Infrastructure 30
3.4 Case Studies (AKRSP, GLOF II, ICIMOD) 33
4. Modern Irrigation Techniques and Efficient Water Use 36
4.1 Overview of Modern Irrigation Techniques 36
4.3 Digital Tools: GIS, Flow Monitoring, Automation 38
4.4 Barriers and Enablers for Adoption 39
4.5 Selecting and Adapting Irrigation Innovations in High-Mountain Contexts 41
5. Operations, Maintenance, and Financing Models 44
5.1 Traditional O&M Practices and Community Norms 44
5.2 Informal Community Systems and External Support Mechanisms 45
5.3 Emerging Cost-Sharing and Financing Models 46
5.4 Public Sector Support and Partnership Approaches 48
5.5 Lessons from Regional and Global Experience for Sustainable O&M 49
5.6 Institutionalization and Professionalization of Operation and Maintenance 50

6. Synthesis of Key Findings, Gaps, and Constraints 52
6.1 Key Findings 52
6.2 Key Gaps and Constraints 53
7. Implementation, Fiscal Context and Strategic Recommendations 54
7.1 Phased Implementation Pathway 54
7.2 Public Financing Context 54
7.3 Thematic Recommendations 55
8. Legal and Policy Reform Agenda 58
8.1 Formal Recognition of Customary Water Management Institutions 58
8.2 Enabling Frameworks for Community Water Management Utility Companies 58
8.3 Clarification of Public Sector Stewardship Roles 58
8.4 Financing and Public Support Instruments 59
8.5 Embedding Social Safeguards and Inclusive Governance 59
8.6 Learning-Oriented Legal Adaptation 59
8.7 Monitoring and Evaluation Framework 59
Annexures 61
Annex A: Literature Review 61
Annex B: Field Instruments and Research Tools 66
Annex C: Summary of Evidence Across all Domains of Enquiry 70
Annex D: Summary of Stakeholder Consultations and Key Inputs 82
Annex E: M&E Framework for the Legal and Policy Reform Agenda 85
Annex F: Alignment Matrix of Irrigation Sector KPIs with SDG Targets 86
Bibliography 88

List of Tables

Table 1. National Water Policies and their Relevance to Irrigation in GB 15
Table 2. Customary Water Management Institutions by Region 19
Table 3. Institutional Mandates and Actual Roles in Irrigation Governance 20
Table 4. Coordination Challenges and Opportunities in Irrigation Governance 24
Table 5. Typology of Irrigation Systems in Gilgit-Baltistan 27
Table 6. Common Failure Patterns and Vulnerabilities in Irrigation Infrastructure 29
Table 7. Climate Stressors, Infrastructure Exposure, and System-Level Impact 32
Table 8. Comparative Analysis of Selected Irrigation Case Studies 34
Table 9. Barriers and Enablers to Adoption of Modern Irrigation Technologies 41
Table 10. Adaptability of Irrigation Technologies in High-Mountain Contexts 42
Table 11. Dominant O&M Arrangements by Region 46
Table 12. Prevalence of Cost-Sharing and Financing Practices (By region) 47
Table 13. Frequency of Reported Constraints to Monetary Cost-Sharing (By region) 48
Table 14. ADP Allocation for the Irrigation Department 55

List of Figures

Figure 1. Map of Gilgit-Baltistan 9
Figure 2. Methodological framework 12
Figure 3. Key stakeholders 13
Figure 4. Key stakeholders 22
Figure 5. Solar Lift-Storage-Gravity Systems 38

Executive Summary

This Irrigation Sector Study was undertaken as part of a Policy Support Technical Assistance project funded by the European Union for the Government of Gilgit-Baltistan. It examines the policy, institutional, technical, and operational foundations of irrigation systems in Gilgit-Baltistan, with a focus on sustainability, resilience, and institutional fit. The study is grounded in extensive field-based evidence, stakeholder consultations, and a review of relevant policy and technical literature. It recognises irrigation as the backbone of agriculture and rural livelihoods in a high-mountain, water-scarce environment, while highlighting emerging pressures linked to climate change, demographic shifts, and institutional gaps. The analysis confirms that future gains in irrigation performance will depend less on infrastructure expansion and more on strengthening community-managed systems through appropriate public stewardship, institutional upgrading, and targeted investment.

A central finding of the study is that irrigation in Gilgit-Baltistan is fundamentally a community-owned and community-managed system. More than 90 percent of traditional irrigation infrastructure has been developed, operated, and maintained through informal customary arrangements that have evolved in response to extreme environmental constraints, including low precipitation, high seasonal variability, and reliance on snowmelt and glacial runoff. Indigenous engineering practices, collective labour mobilisation, and customary governance institutions have enabled sustained agricultural production in an otherwise fragile and inhospitable environment.

The assessment finds that, while recent legislative and policy developments signal growing recognition of community roles and environmental protection, formal institutional engagement with irrigation management remains limited. Public sector involvement is largely confined to capital investments, post-disaster rehabilitation, and project-based interventions, with minimal presence in routine operation and maintenance. This has resulted in a persistent disconnect between formal policy frameworks and the informal systems that continue to dominate irrigation management on the ground.

Infrastructure conditions across the region are deteriorating. Ageing channels, deferred maintenance, and increasing exposure to climate-related hazards—particularly floods, landslides, and glacial lake outburst floods—are placing growing stress on irrigation systems. While communities continue to mobilise labour and local resources for routine upkeep, the scale and technical complexity of damage increasingly exceed local capacity, leading to episodic dependence on external assistance and heightened vulnerability of agricultural livelihoods.

Analysis of modern irrigation technologies indicates that limited adoption is driven less by lack of awareness than by institutional and economic constraints. In collectively managed systems governed by customary allocation rules, individual investments in efficiency-enhancing technologies yield limited private returns. Combined with subsistence-oriented cropping patterns and the availability of subsidised staple food, this weakens incentives for farmers to invest in capital-intensive technologies. Where such technologies have been introduced, uptake has generally remained partial, project-bound, and dependent on continued external support.

Operation and maintenance systems based on customary labour remain functional but are under increasing strain. Labour shortages linked to migration and livelihood diversification, rising cash requirements for technically complex repairs, and escalating climate risks are eroding the viability of purely voluntary arrangements. At the same time, field evidence and comparative experience suggest that abrupt formalisation or externally imposed management models are poorly suited

to the Gilgit-Baltistan context. More sustainable outcomes are associated with incremental strengthening of existing institutions rather than their replacement.

The study identifies several structural constraints that limit long-term sustainability. These include the absence of systematic empirical data on system performance and maintenance costs; fragmented and weakly coordinated institutional mandates; structurally constrained financing arrangements for routine operation and maintenance; and a persistent gap between customary governance systems and formal institutional frameworks. Technology-led solutions, in the absence of enabling institutional and incentive structures, are unlikely to deliver transformational improvements at scale.

Taken together, the findings underscore that the future sustainability of irrigation in Gilgit-Baltistan depends not on wholesale system transformation, but on carefully sequenced, institutionally grounded interventions. Strengthening irrigation performance will require approaches that recognise the centrality of community-managed systems, address emerging technical and financial constraints, and gradually bridge customary practices with formal policy and support mechanisms. These conclusions provide the analytical foundation for the strategic options and recommendations set out in the subsequent section of the report.

List of Acronyms

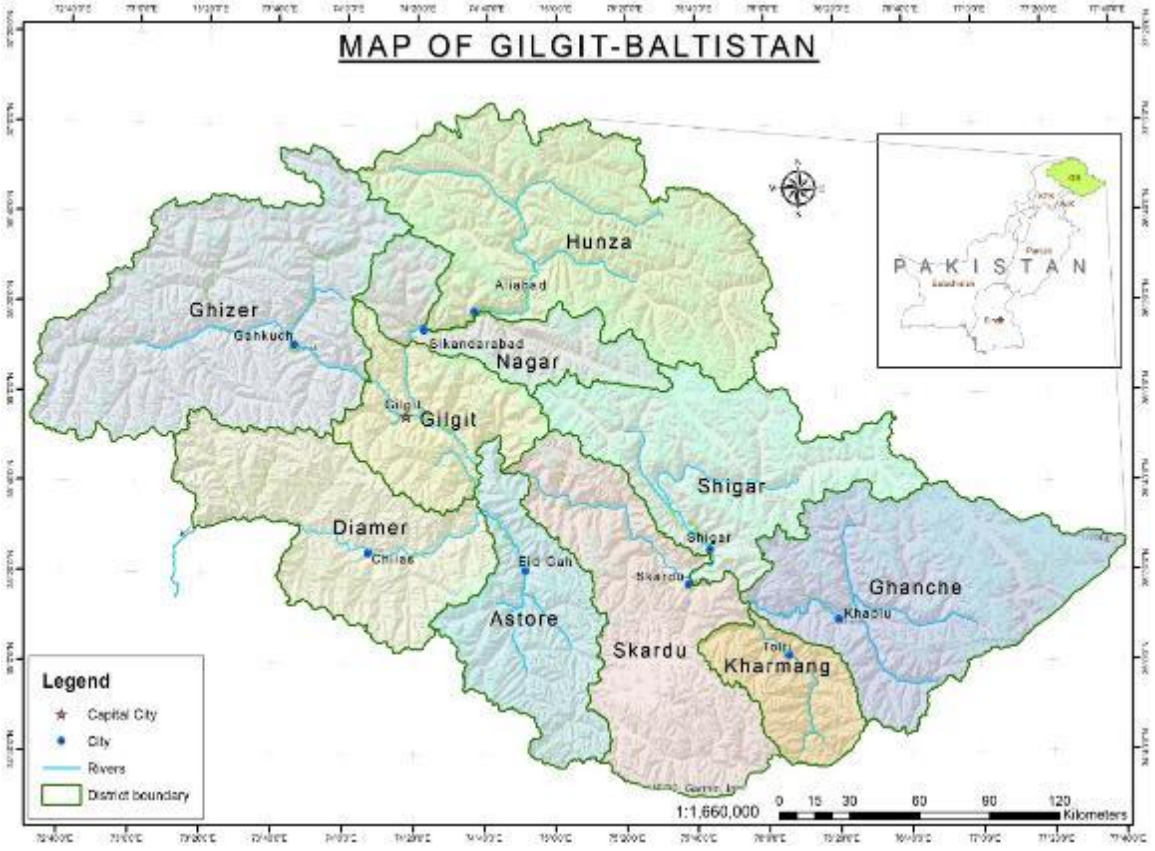
AKDN	Aga Khan Development Network
AKRSP	Aga Khan Rural Support Programme
AR6	Sixth Assessment Report (IPCC)
EU	European Union
EUD	European Union Delegation
GB	Gilgit-Baltistan
GBDMA	Gilgit-Baltistan Disaster Management Authority
GBEPA	Gilgit-Baltistan Environmental Protection Agency
GBRSP	Gilgit-Baltistan Rural Support Programme
GHGs	Greenhouse Gases
GIS	Geographic Information System
GLOF	Glacial Lake Outburst Flood
GoGB	Government of Gilgit-Baltistan
GoP	Government of Pakistan
HKH	Hindu Kush–Karakoram–Himalayan
ICIMOD	International Centre for Integrated Mountain Development
IFAD	International Fund for Agricultural Development
IPCC	Intergovernmental Panel on Climate Change
LG&RD	Local Government and Rural Development Department
NGO	Non-Governmental Organization
NRM	Natural Resource Management
NTU	NTU International
O&M	Operation and Maintenance
OECD	Organisation for Economic Co-operation and Development
DAC	Development Assistance Committee
P&D	Planning and Development Department
SDGs	Sustainable Development Goals
TA	Technical Assistance
UNDP	United Nations Development Programme
WUA	Water User Association

1. Introduction

1.1 The Agroecological Context

Gilgit-Baltistan is characterised by an extreme high-mountain environment shaped by a partial rain-shadow effect, very low annual precipitation, and heavy reliance on cryosphere water sources. Rain-fed agriculture is virtually absent, and crop production depends almost entirely on irrigation derived from snowmelt and glacial runoff. The spatial and seasonal mismatch between water availability and cultivable land has historically required complex diversion systems to transport water from high-altitude sources to valley floors. As documented in regional mountain studies, this environmental constraint makes irrigation not merely an agricultural input but the foundational condition for human settlement and food production in the region (Kreutzmann, Irrigation and Water Management in the Karakoram, 2011).

Figure 1. Map of Gilgit-Baltistan



Map source: GB Environment, forest and wildlife department

In response to these constraints, indigenous communities developed extensive gravity-fed irrigation networks, often cutting channels across sheer rock faces and unstable slopes using simple hand tools and locally available materials. These systems were constructed, operated, and maintained collectively over generations, reflecting a high degree of local engineering knowledge and social organisation. Alongside physical infrastructure, communities evolved customary water management arrangements that governed allocation, seasonal scheduling, conflict resolution, and maintenance responsibilities. Roles such as locally recognised water managers, elders' councils, and collective labour institutions emerged to regulate access to water, enforce compliance, and mobilise labour for desilting and repairs. Maintenance obligations were

commonly met through pooled labour and, in some cases, contributions in kind, reinforcing social cohesion while sustaining system functionality under conditions of limited cash availability (IUCN Pakistan, Customary Water Management Practices in Northern Pakistan, 2000; Kreutzmann, 2011).

Over time, these long-standing systems have come under increasing stress. Economic diversification, expanded access to education, and labour outmigration—particularly among younger cohorts—have reduced the availability of voluntary labour that traditionally underpinned operation and maintenance. At the same time, irrigation infrastructure is ageing, technical demands are increasing, and climate-related hazards such as floods, landslides, and glacial lake outburst floods are occurring with greater frequency and intensity. Primary evidence reviewed under this study indicates that more than 90 percent of traditional irrigation infrastructure in Gilgit-Baltistan remains community-owned and managed through informal customary arrangements, with minimal routine involvement of public sector institutions. While these systems continue to demonstrate resilience, the combined pressures of climate risk, demographic change, and limited financing capacity have exposed structural vulnerabilities. This evolving context provides a clear rationale for the present irrigation sector study, which seeks to examine how indigenous, community-managed systems can be strengthened and supported through appropriate policy, institutional, and technical interventions without undermining their local legitimacy or social foundations (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024; IUCN Pakistan, 2000).

1.2 Background of the TA Assignment

This Irrigation Sector Study has been prepared under the EU-funded Policy Support Technical Assistance (TA) to the Government of Gilgit-Baltistan, which aims to strengthen evidence-based policy formulation and institutional capacity for climate resilience and green development across priority sectors. The Technical Assistance supports the Government of Gilgit-Baltistan in developing, refining, and implementing policies related to irrigation, agriculture, energy, disaster risk reduction, and climate action over a 20-month implementation period between 2024 and 2026 (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, Version 1.02, 2024).

The Technical Assistance is designed to provide analytical inputs, institutional diagnostics, and practical policy advice aligned with EU development cooperation priorities, including climate action, environmental sustainability, gender equality, and inclusion of vulnerable and marginalised groups, including persons with disabilities (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, Version 1.02, 2024).

Within this broader framework, irrigation has been identified as a critical sector due to its central role in agriculture, food security, and settlement in Gilgit-Baltistan. Irrigation systems in the region are predominantly gravity-fed and community-managed, having evolved through long-standing customary arrangements and collective labour-based construction and maintenance practices. Primary field evidence indicates that water allocation, routine operation, and maintenance are governed through locally recognised institutions rather than formal public service delivery mechanisms (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

The strategic focus on irrigation is also shaped by national water policy priorities. The National Water Policy of Pakistan establishes a hierarchy of water use that prioritises drinking water, followed by irrigation, energy generation, and ecological flows. Translating these priorities into

operational practice in Gilgit-Baltistan remains challenging due to climate variability, fragile infrastructure, and limited institutional coordination at the provincial and local levels (National Water Policy of Pakistan, 2018; Field Work Strategy and Data Collection Instruments for Irrigation Sector Study, EU Policy Support TA, August 2025).

1.3 Objectives of the Irrigation Sector Study

The irrigation sector study is intended to provide an evidence-based assessment of the irrigation sector in Gilgit-Baltistan, focusing on identifying structural constraints, vulnerabilities, and realistic pathways for improving system performance and resilience. The study is designed as a policy- and institutionally oriented analysis rather than a technical engineering audit.

The objectives of the irrigation component include assessing existing policy, legal, and institutional frameworks governing irrigation; analysing infrastructure condition, reliability, and exposure to climate stress; examining the feasibility and limitations of modern irrigation techniques such as drip and sprinkler systems; reviewing operation and maintenance practices and financing arrangements; and developing performance indicators aligned with national priorities, the Sustainable Development Goals, and EU policy frameworks (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

1.4 Methodology and Approach

The study adopts a qualitative, evidence-led analytical approach that integrates structured primary field evidence with a targeted review of policy documents, legislation, programme materials, and relevant academic and grey literature. The approach is designed to generate policy-relevant insights into irrigation governance, infrastructure performance, and institutional arrangements in Gilgit-Baltistan, while remaining grounded in observed practices and lived realities.

Primary evidence was generated through structured stakeholder consultations conducted across multiple regions of Gilgit-Baltistan, including Gilgit, Baltistan, and Diamer/Astore. Consultations engaged farmers, customary water managers, community representatives, and relevant government institutions. Data collection instruments were designed to capture institutional roles, operation and maintenance practices, financing arrangements, equity considerations, and exposure to climate-related risks. Responses were recorded, expanded, and coded using a structured framework that links each response to a specific question, stakeholder type, region, and reference code. This framework is consolidated in Annex C, enabling traceability from synthesized findings to full transcripts and field notes.

The fieldwork approach emphasised triangulation across multiple sources of evidence. Primary consultation findings were cross-checked against regional patterns, stakeholder perspectives, and secondary documentation to identify convergence, divergence, and contextual variation. Internal quality control measures were applied throughout the process, including consistency checks across transcripts, comparative review of regional findings, and iterative validation of emerging themes against the original consultation instruments. These measures were intended to strengthen analytical robustness and reduce the risk of anecdotal or isolated interpretations (Field Work Strategy and Data Collection Instruments for Irrigation Sector Study, EU Policy Support TA, August 2025).

Secondary analysis draws on documents provided under the Technical Assistance, including policy frameworks, legislation, programme reports, and technical studies relevant to irrigation and water management in Gilgit-Baltistan. In addition, a consolidated literature review was

undertaken to situate the field findings within broader national and international experience on community-managed irrigation systems, climate resilience, institutional reform, and public–community interfaces. The literature review is used to contextualise and interpret primary evidence rather than to substitute for it, recognising the limited availability of region-specific empirical research on irrigation in Gilgit-Baltistan (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Overall, the methodology prioritises analytical coherence, contextual sensitivity, and policy relevance. It is explicitly designed to surface institutional dynamics, incentives, and constraints that shape irrigation performance, rather than to generate statistically representative estimates or engineering-level assessments.

Figure 2. Methodological framework



1.5 Stakeholder Analysis

Stakeholder analysis formed an integral part of the study's methodological approach and informed both data collection and interpretation. Stakeholders were identified and engaged based on their roles across the irrigation system lifecycle, including planning, construction, operation, maintenance, financing, regulation, and external support.

At the community level, the study engaged farmers, customary water managers (including elders, mirabs, and water committees), and other local representatives directly involved in water allocation and maintenance. These stakeholders provided insight into day-to-day irrigation practices, collective decision-making processes, conflict resolution mechanisms, and the practical challenges associated with system upkeep and climate risk.

At the institutional level, consultations included relevant government departments and agencies with mandates related to irrigation, water resources, agriculture, local government, and disaster

response. Engagement focused on understanding formal roles, coordination mechanisms, funding arrangements, and interfaces with community-managed systems. Development partners and non-governmental organisations with experience in irrigation and rural development were also considered where relevant, particularly in relation to project-based interventions and capacity-building efforts.

Stakeholder perspectives were analysed comparatively to identify areas of alignment and divergence between customary systems and formal governance frameworks. This analysis helped to clarify institutional gaps, overlapping mandates, and opportunities for more effective collaboration. It also informed the identification of feasible entry points for reform that build on existing roles and capacities rather than imposing externally defined institutional models.

Figure 3. Key stakeholders



1.6 Limitations and Scope

The assessment covers Gilgit-Baltistan as a whole, drawing on evidence from multiple regions to reflect variation in hydrological, geomorphological, and institutional conditions. For analytical purposes, regional groupings—Gilgit, Baltistan, and Diamer/Astore—are used to distinguish relatively stable valley systems from more fragile upper catchments that are exposed to recurrent climate-related hazards and infrastructure shocks.

The analytical scope includes irrigation governance and institutional arrangements, infrastructure condition and operation, water allocation and equity considerations, exposure to climate risks, and the role of external support and innovation. The study does not undertake detailed hydrological modelling, engineering design verification, or quantitative water balance analysis, nor does it assess individual irrigation schemes at design level. Financial analysis is similarly qualitative in nature and focuses on institutional arrangements and feasibility rather than on cost modelling.

These limitations are intentional and reflect the study's focus on system-level performance and policy-relevant insights derived from primary field evidence and targeted secondary review. The findings are intended to inform strategic decision-making, institutional strengthening, and the design of future interventions rather than to replace detailed technical or engineering studies (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

2. Policy and Institutional Framework

This section reviews the national and provincial policy and institutional framework governing irrigation in Gilgit-Baltistan, with a focus on how higher-level policy intent translates into operational realities in a high-mountain, community-managed irrigation context. The analysis under TOR 1 examines alignment, applicability, and gaps between national policy directions and the institutional arrangements and practices observed in Gilgit-Baltistan, drawing on policy documents, technical strategies, and primary field evidence.

At the national level, irrigation policy is shaped primarily by overarching water governance frameworks rather than irrigation-specific legislation tailored to mountain regions. These national policies provide strategic direction on water allocation priorities, conservation, efficiency, and climate resilience, but leave significant discretion to provincial and regional governments in terms of implementation. Understanding the scope and limitations of these national frameworks is therefore essential for assessing their relevance to Gilgit-Baltistan's irrigation systems. Key areas of alignment, omission, and limited applicability are summarised in Table 1.

Table 1. National Water Policies and their Relevance to Irrigation in GB

Policy / Strategy	Year	Irrigation Focus	Treatment of Community Systems	Relevance to GB Context	Key Gaps for GB
National Water Policy	2018	Integrated water resources management; conservation and efficiency; agriculture as major water user; institutional strengthening.	Community-managed irrigation systems not explicitly recognised; assumes formal irrigation agencies.	Limited relevance to glacier-fed, small-scale gravity systems typical of GB.	Does not address customary O&M arrangements or informal collective governance central to irrigation in GB *(Field consultations; Annex C).*
National Water Conservation Strategy	2023–27	Promotion of efficient water use; irrigation efficiency technologies; behavioural change; coordination across sectors.	Community participation referenced in general terms, without operational guidance for customary systems.	Does not differentiate mountain irrigation from plains-based systems.	Technology-centric measures not aligned with subsistence agriculture and collective irrigation management observed in GB *(Field consultations; Annex C).*

National Climate Change Policy	2021	Climate resilience; adaptation in water and agriculture; recognition of glacier melt and flood risks.	Community-based adaptation acknowledged, but not linked to irrigation institutions.	Glacier dynamics recognised, but irrigation infrastructure in high-mountain contexts not addressed in detail.	Lacks guidance on strengthening community-managed irrigation systems most exposed to climate risks in GB *(Field consultations; Annex C).*
Other federal water and agriculture strategies	Various	Productivity enhancement; modern irrigation; efficiency improvements.	Framed around formal programmes and project-based delivery.	Limited agro-ecological differentiation.	Institutional and incentive constraints highlighted in GB are not addressed (*Field consultations; Annex C; Annex A).*

2.1 National Policies

Pakistan's National Water Policy (2018) provides the primary national framework guiding water resource management across sectors, including irrigation. The policy establishes a hierarchy of water use that prioritises drinking water, followed by irrigation, energy generation, and environmental flows. It recognises water scarcity, climate variability, and inefficiencies in water use as central challenges and calls for improved governance, conservation, and demand management across the water sector (National Water Policy of Pakistan, 2018).

In relation to irrigation, the National Water Policy emphasises improving water use efficiency, reducing conveyance losses, promoting modern irrigation technologies, and strengthening institutional coordination across federal and provincial levels. The policy also acknowledges the role of community participation in water management, though this is framed largely in general terms rather than through explicit recognition of customary irrigation institutions that dominate in regions such as Gilgit-Baltistan (National Water Policy of Pakistan, 2018).

The policy's relevance to Gilgit-Baltistan is therefore mixed. While its emphasis on climate resilience, conservation, and efficient water use aligns with the region's growing exposure to climate-induced risks, the policy does not differentiate between lowland canal systems and gravity-fed, community-managed irrigation systems typical of high-mountain environments. Primary field evidence from Gilgit-Baltistan indicates that many of the policy's proposed efficiency measures—such as widespread adoption of high-efficiency irrigation systems—are constrained by infrastructural fragility, seasonal water availability, and collective governance arrangements (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

The National Water Conservation Strategy (NWCS) 2023–27 builds on the National Water Policy by translating broad policy objectives into a time-bound strategic framework focused on water conservation, efficiency, and behavioural change. The strategy places strong emphasis on reducing water losses, promoting efficient irrigation practices, improving monitoring and data

systems, and raising awareness among water users. It also highlights the importance of climate adaptation and resilience in water management planning (National Water Conservation Strategy of Pakistan, 2023–2027).

From the perspective of irrigation in Gilgit-Baltistan, the NWCS reinforces national priorities around efficiency and conservation but remains largely generic in its treatment of regional diversity. The strategy does not provide specific guidance on how conservation and efficiency objectives should be operationalised in remote, mountainous regions where irrigation systems are small-scale, community-financed, and highly exposed to natural hazards. Field evidence suggests that without contextual adaptation, conservation-oriented strategies risk placing additional expectations on communities without addressing upstream infrastructure vulnerabilities or institutional capacity constraints (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Both the National Water Policy and the National Water Conservation Strategy articulate ambitious national goals for improving irrigation efficiency and sustainability. However, their practical influence in Gilgit-Baltistan is mediated by the absence of region-specific implementation frameworks and limited integration with customary irrigation institutions. The literature review notes that national water strategies in Pakistan have historically been more effective in informing large-scale irrigation planning than in shaping outcomes in marginal and mountain regions, where local governance structures and physical constraints play a dominant role (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

In summary, national water policies provide an important strategic backdrop for irrigation reform, but their applicability to Gilgit-Baltistan depends on contextual interpretation and institutional adaptation. The following sections examine how these national directions interact with provincial frameworks, legal arrangements, and customary institutions at the regional level.

2.2 Provincial Policies

At the provincial level, irrigation governance in Gilgit-Baltistan is shaped by a combination of sectoral policies, administrative arrangements, and evolving legal recognition of community-based institutions. Unlike Pakistan's provinces, Gilgit-Baltistan operates under a distinct constitutional and administrative framework, which affects how national policies are interpreted and implemented in practice. Provincial policies relevant to irrigation therefore tend to be embedded within broader development, agriculture, and climate-related strategies rather than articulated through a standalone irrigation policy.

Institutionally, irrigation responsibilities in Gilgit-Baltistan are distributed across multiple government departments, with primary roles played by the Irrigation Department, Agriculture Department, Planning and Development Department, and district administrations. These institutions are responsible for planning, rehabilitation, and oversight of irrigation infrastructure, particularly for schemes that receive public funding. However, their role in routine operation and maintenance is limited, as most irrigation systems remain under community management once constructed or rehabilitated (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, Version 1.02, 2024).

Primary field evidence indicates that this division of responsibilities is well understood at the community level but creates gaps in coordination and accountability at the system level. Communities retain responsibility for day-to-day operation, maintenance, and water allocation, while government engagement is typically episodic and reactive, focused on capital works or post-

disaster rehabilitation rather than preventive maintenance or performance monitoring (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Provincial development strategies and sector plans recognise irrigation as a foundational input for agriculture and rural livelihoods, but they generally frame irrigation investment in terms of infrastructure expansion and rehabilitation rather than institutional strengthening. The absence of a dedicated provincial irrigation policy limits the articulation of clear service standards, maintenance responsibilities, or financing mechanisms tailored to community-managed systems. This contributes to a reliance on project-based interventions rather than sustained institutional support.

Governance arrangements at the local level are dominated by customary institutions that regulate water distribution, labour mobilisation, and conflict resolution. These institutions operate alongside formal government structures but are not systematically integrated into provincial planning or budgeting processes. Field evidence suggests that while customary systems are effective in managing routine operations, they lack formal channels to influence investment priorities, access technical support, or secure predictable financing for major repairs (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Recent legislative developments in Gilgit-Baltistan have begun to recognise community organisations as legitimate actors in local governance, including in natural resource management. However, the literature review notes that such recognition has yet to translate into clearly defined roles for irrigation institutions within provincial administrative systems. As a result, customary irrigation governance remains largely informal from a policy perspective, despite its central role in sustaining irrigation services (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Overall, provincial policies and institutional arrangements in Gilgit-Baltistan provide a partial and fragmented framework for irrigation governance. While government institutions play a critical role in capital investment and post-disaster response, long-term system performance depends heavily on community institutions that operate with limited formal support. This misalignment between institutional responsibility and authority represents a central governance challenge under TOR 1 and shapes the effectiveness of irrigation policy implementation in the region.

2.3 Customary Law and Indigenous Institutions

Irrigation systems in Gilgit-Baltistan are predominantly governed through customary law and indigenous institutional arrangements that have evolved over generations in response to extreme environmental constraints. In a context characterised by aridity, steep terrain, and dependence on snowmelt and glacial water, local communities developed irrigation channels through collective effort, often cutting through rock faces and unstable slopes using simple tools. These physical systems were accompanied by equally robust social institutions that regulate water allocation, organise operation and maintenance, resolve disputes, and mobilise labour and resources.

Across all regions of Gilgit-Baltistan, customary irrigation institutions operate on the basis of locally recognised authority, shared norms, and collective decision-making rather than formal legal status. Water allocation rules are typically unwritten but widely understood, enforced through social consensus and mediated by designated custodians or committees. These arrangements have historically ensured equitable distribution of water, coordinated maintenance activities, and managed conflicts in settings where state presence was minimal or absent.

Field consultations confirm that these indigenous institutions continue to bear primary responsibility for the operation and maintenance of most traditional irrigation systems. However, they are under increasing stress due to climate variability, more frequent damage from floods and glacial lake outburst events, declining availability of communal labour, and socio-economic change, including outmigration and diversification of livelihoods. Despite their central role, these institutions remain largely unrecognised within formal policy and legal frameworks, limiting their access to technical support, public financing, and institutional strengthening measures.

The nature and expression of customary irrigation institutions vary by region, reflecting differences in social organisation, ecological conditions, and historical practice. Table 2 summarises the principal indigenous water management institutions identified through field consultations across Gilgit, Baltistan, and Diamer/Astore.

Table 2. Customary Water Management Institutions by Region

Region	**Local Term(s) for Water Management**	**Core Functions**	**Governance Logic and Authority**	**Role in Operation and Maintenance**	**Key Pressures Identified**
Gilgit Region	Mirab (water manager); informal water committees	Allocation and scheduling of water; coordination of maintenance; mobilisation of communal labour; dispute resolution	Authority derived from customary norms, community consensus, and elder mediation	Central role in day-to-day operation and maintenance of gravity-fed irrigation systems	Declining communal labour; ageing infrastructure; increased climate-related damage; lack of formal recognition (Field consultations; Annex C).
Baltistan Region	Mirab or locally recognised water custodians	Regulation of water distribution; maintenance of channels and intakes; coordination of repair works; conflict management	Strong customary norms embedded in village-level collective decision-making	Continuous responsibility for operation and maintenance across most traditional systems	High exposure to floods and GLOFs; growing maintenance burden; limited technical and financial support (Field consultations; Annex C).
Diamer / Astore Region	Locally recognised water custodians and informal committees (terminology varies by valley)	Water distribution; upkeep of diversion structures; mobilisation of labour; conflict resolution	Informal leadership structures and customary rules enforced through collective agreement	Primary responsibility for operation and maintenance of remote and climate-exposed systems	Climate variability; remoteness; weak linkages with public institutions (Field consultations; Annex C).

Source: Field consultations and coded summaries, Annex C; regional consultation transcripts – Gilgit, Baltistan, Diamer/Astore)

2.4 Institutional Landscape: Roles and Mandates

The institutional landscape governing irrigation in Gilgit-Baltistan is characterised by a multiplicity of actors with partially defined and weakly coordinated roles. Responsibilities for policy formulation, planning, infrastructure development, and service delivery are distributed across provincial departments, district administrations, and community institutions, with no single entity exercising comprehensive stewardship over the irrigation sector.

At the provincial level, the Irrigation Department holds the formal mandate for planning, construction, and rehabilitation of irrigation infrastructure financed through public funds. Its role is largely concentrated on capital works, including new schemes and post-disaster rehabilitation, rather than routine operation or maintenance. Once infrastructure is completed or restored, responsibility for day-to-day operation and maintenance is typically transferred back to local communities, with limited follow-up or performance monitoring by the department (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

The divergence between formal institutional mandates and the roles institutions perform in actual practice is summarised in Table 3, highlighting the gap between statutory responsibilities and the de facto functions observed through field evidence and stakeholder consultations.

Table 3. Institutional Mandates and Actual Roles in Irrigation Governance

Institution / Actor	Formal Mandate (as per rules / policy)	Actual Role Observed in Practice	Nature of Engagement	Key Gaps Identified
Irrigation Department (GB)	Planning, design, and construction of irrigation infrastructure; oversight of irrigation assets.	Primarily involved in construction and episodic rehabilitation of selected schemes; limited engagement in routine operation and maintenance of community systems.	Project-based and episodic.	Limited mandate and capacity for supporting O&M of community-managed systems; weak interface with customary institutions (Field consultations; Annex C).
Local Government & Rural Development Department (LG&RD)	Support to local governance structures; implementation of local development schemes.	Occasional support through small grants or rehabilitation works, particularly following disaster events; no systematic role in irrigation O&M.	Ad hoc and reactive.	Absence of a clear stewardship role for irrigation institutions at local level; limited regulatory or facilitative engagement (Field consultations; Annex C).
Agriculture Department	Promotion of agricultural productivity, extension services,	Engagement largely limited to crop advisory services; weak integration with irrigation	Sector-specific and disconnected.	Limited coordination between water management and agricultural extension despite

	and crop development.	management and water allocation practices.		interdependence (Field consultations; Annex C).
Disaster Management Authorities	Preparedness and response to natural disasters, including floods and GLOFs.	Involved mainly in post-disaster assessments and emergency response; limited role in preventive strengthening of irrigation systems.	Event-driven.	Focus on response rather than resilience; weak linkage with irrigation institutions managing climate-exposed assets (Field consultations; Annex C).
Customary / Community Irrigation Institutions	No formal legal mandate under statutory law.	Responsible for water allocation, operation, maintenance, conflict resolution, and mobilisation of labour and contributions across most traditional irrigation systems.	Continuous and system-wide.	High institutional burden without legal recognition, technical support, or predictable financing; increasing stress due to social and climatic change (Field consultations; Annex C).

The Agriculture Department plays an indirect role in irrigation governance, primarily through extension services, crop advisory functions, and promotion of agricultural productivity. While irrigation is recognised as a critical input for agriculture, coordination between agricultural extension and irrigation management is limited. Field evidence suggests that irrigation considerations are rarely integrated into extension advice or cropping recommendations, resulting in a disconnect between water availability, cropping patterns, and productivity initiatives (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA).

The Planning and Development Department exercises a central role in allocating development budgets and approving projects, including those related to irrigation infrastructure. However, its involvement is primarily procedural and financial, with limited engagement in sector-specific performance oversight. As a result, irrigation investments are often assessed in terms of expenditure and physical outputs rather than long-term functionality or service reliability.

At the district and sub-district levels, administrative authorities facilitate implementation and coordination but do not typically hold technical mandates related to irrigation management. Their involvement is most visible during emergency situations, such as flood damage or infrastructure failure, when they act as intermediaries between communities and provincial departments. Outside such events, district administrations have limited capacity or authority to influence irrigation system performance (Annex C: Summary of Primary Evidence from the Field).

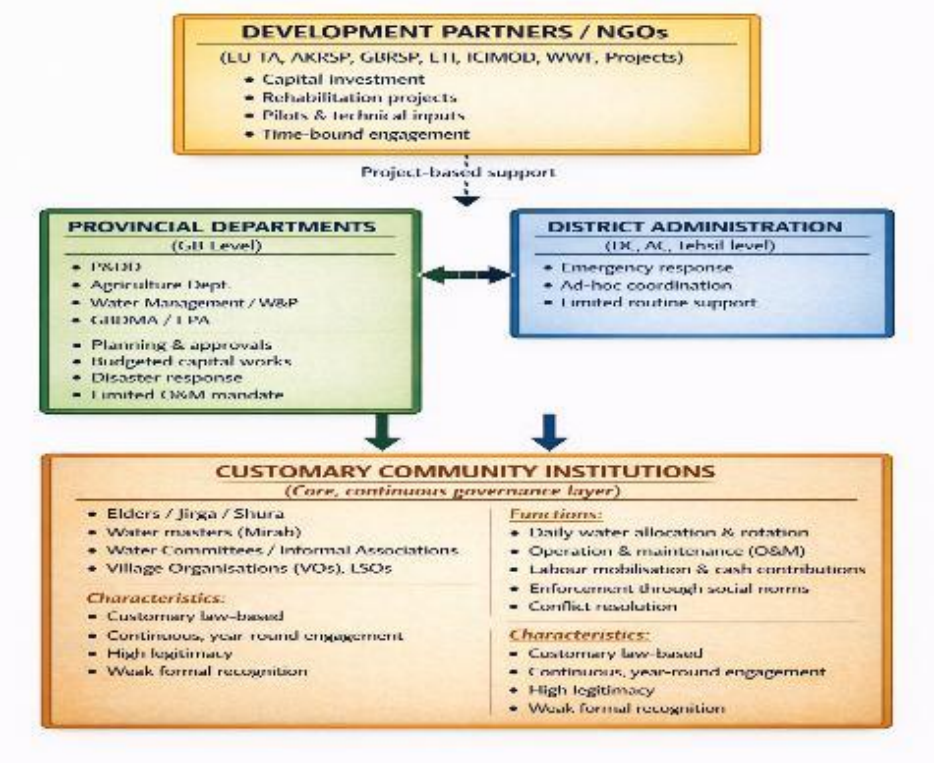
Community institutions constitute the most operationally significant actors in the irrigation landscape. Customary irrigation institutions manage water allocation, enforce rules, mobilise labour for maintenance, and resolve disputes. These institutions operate continuously and bear the primary responsibility for system performance, yet they lack formal recognition within provincial administrative systems. Their mandates are socially defined rather than legally codified, limiting their ability to access public resources or participate in formal decision-making processes

(Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Non-governmental organisations and development projects have intermittently influenced the institutional landscape through rehabilitation works, pilots, and capacity-building initiatives. While such interventions have introduced technical innovations and temporary institutional arrangements, their roles are time-bound and not embedded within permanent governance structures. Field evidence indicates that once project support ends, institutional responsibilities revert to existing community and government actors without sustained mechanisms for coordination (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Overall, the institutional landscape for irrigation in Gilgit-Baltistan reflects a clear imbalance between responsibility and authority. Communities are responsible for daily operation and maintenance but lack formal mandates and resources, while government institutions hold formal authority but engage primarily at the level of infrastructure investment and crisis response. This fragmentation constrains accountability, weakens performance management, and limits the effectiveness of policy implementation. Clarifying roles and strengthening coordination across institutions remains a central challenge and sets the context for the subsequent discussion on gaps, overlaps, and coordination challenges (Figure 4).

Figure 4. Key stakeholders



2.5 Coordination Challenges and Opportunities

Coordination across policy, institutional, and operational levels represents one of the most persistent challenges in irrigation governance in Gilgit-Baltistan. While national policies articulate strategic objectives for water conservation, efficiency, and climate resilience, and provincial institutions hold formal mandates for planning and investment, day-to-day irrigation performance

is shaped primarily by community-managed systems operating with limited institutional linkage to higher-level decision-making.

Primary field evidence indicates that coordination between provincial departments involved in irrigation, agriculture, planning, and disaster response is largely ad hoc. Interactions tend to occur in response to infrastructure failure or extreme events rather than through routine planning or performance monitoring processes. As a result, irrigation systems are often addressed as isolated infrastructure assets rather than as part of an integrated water–agriculture–livelihood system (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

A significant coordination gap exists between irrigation management and agricultural extension services. Field evidence suggests that cropping recommendations, input promotion, and productivity initiatives are rarely aligned with actual water availability or irrigation reliability. This disconnect limits the effectiveness of agricultural support programmes and constrains the potential for irrigation investments to translate into sustained productivity gains (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Coordination between government institutions and customary irrigation bodies is also limited. Although community institutions are responsible for routine operation and maintenance, they are not formally integrated into provincial planning, budgeting, or monitoring frameworks. This results in a structural asymmetry in which communities bear operational responsibility without corresponding authority or access to technical and financial support. Field evidence indicates that communication between communities and government departments typically occurs through informal or political channels, reducing transparency and predictability (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, Version 1.02, 2024).

The literature review highlights that such coordination failures are common in regions where irrigation systems are small-scale, dispersed, and community-managed. Without explicit institutional mechanisms to link local governance structures with provincial planning and national policy frameworks, sector performance remains reactive and fragmented. Studies emphasise that improving coordination does not necessarily require new institutions, but clearer interfaces, role definition, and information flows across existing actors (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Despite these challenges, field evidence also points to coordination opportunities. Customary irrigation institutions demonstrate strong capacity for collective action and local enforcement, which could serve as entry points for improved information sharing, participatory planning, and performance feedback. Similarly, provincial departments involved in planning and disaster response already interact with communities during rehabilitation efforts, providing a potential platform for more systematic engagement if formalised (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

National policy frameworks, including the National Water Policy and the National Water Conservation Strategy, offer additional opportunities by providing a common strategic language around conservation, efficiency, and resilience. Aligning provincial programmes and community-level practices with these shared objectives could improve coherence, provided that implementation approaches are adapted to local institutional and environmental realities (National Water Policy of Pakistan, 2018; National Water Conservation Strategy of Pakistan, 2023–2027).

In summary, coordination challenges in Gilgit-Baltistan stem less from absence of institutions than from weak linkages among them. Opportunities for improvement lie in strengthening interfaces between community institutions and provincial departments, aligning irrigation management with agricultural and disaster risk planning, and using national policy frameworks as guiding references rather than rigid templates. Addressing these coordination issues is central to improving irrigation system performance and sets the foundation for subsequent analysis of infrastructure, resilience, and operational frameworks.

The coordination challenges and opportunities across policy, institutional, and operational levels discussed above are synthesised in Table 4.

Table 4. Coordination Challenges and Opportunities in Irrigation Governance

Coordination Interface	**Intended Coordination (Policy / Mandate)**	**Coordination Observed in Practice**	**Key Challenges Identified**	**Opportunities for Improvement**
Provincial Departments – Community Institutions	Public agencies to plan, construct, and support irrigation infrastructure in coordination with local users.	Interaction largely limited to project implementation and handover; minimal engagement during routine operation and maintenance.	Absence of structured post-handover support; lack of formal recognition of customary institutions; weak feedback loops.	Formalising engagement mechanisms focused on O&M support and institutional strengthening rather than project delivery (Field consultations; Annex C).
Irrigation Department – LG&RD / District Administrations	Complementary roles in infrastructure development and local service delivery.	Coordination is ad hoc and largely event-driven, particularly following natural disasters.	Unclear division of responsibilities; no designated stewardship role at local level for irrigation systems.	Clarifying stewardship and facilitation roles for local governments in supporting community-managed systems (Field consultations; Annex C).
Agriculture Department – Irrigation Institutions	Integrated management of water and agricultural productivity.	Limited coordination between water allocation practices and agricultural extension services.	Sectoral silos; irrigation managed independently of crop planning and extension advice.	Strengthening linkages between irrigation management and agricultural extension, particularly for crop diversification and water efficiency (Field consultations; Annex C).
Disaster Management Authorities – Irrigation Actors	Risk reduction, preparedness, and recovery for	Engagement concentrated on post-disaster response and	Limited preventive integration of irrigation systems into	Incorporating irrigation infrastructure into local resilience and preparedness planning, especially for GLOF- and

	climate-exposed infrastructure.	damage assessment.	disaster risk management frameworks.	flood-prone systems (Field consultations; Annex C).
Customary Institutions – Formal Policy Frameworks	Alignment of local practices with statutory and policy frameworks.	Customary institutions operate largely outside formal legal recognition.	Legal invisibility limits access to public finance, technical support, and institutional upgrading.	Creating enabling legal and policy space for hybrid institutional models that recognise customary governance while introducing professional support functions (Field consultations; Annex C).

Source: Field consultations and coded summaries, Annex C; EU Policy Support Technical Assistance to Government of Gilgit-Baltistan – Inception Report, Version 1.02, 2024)

3. Infrastructure Quality and Design Review

This section reviews the quality, design characteristics, and functional performance of irrigation infrastructure in Gilgit-Baltistan, with a focus on how different system types respond to environmental stress, operational demands, and evolving policy objectives. The analysis under TOR 2 examines infrastructure not as isolated physical assets, but as systems embedded within specific geomorphological, hydrological, and institutional contexts.

Irrigation infrastructure in Gilgit-Baltistan reflects centuries of adaptation to high-mountain conditions, including steep gradients, seasonal meltwater flows, and exposure to floods, landslides, and sediment movement. While recent decades have seen the introduction of externally supported rehabilitation and modernisation efforts, the core design logic of most systems remains rooted in traditional gravity-fed principles. Understanding the typology of irrigation systems is therefore a prerequisite for assessing infrastructure quality, vulnerability, and potential pathways for improvement.

3.1 Typology of Irrigation Systems

Primary field evidence indicates that irrigation systems in Gilgit-Baltistan can be broadly categorised into traditional gravity-fed systems and limited modern or improved variants introduced through public or project-based interventions. Traditional systems dominate both in number and in irrigated area, while modern systems remain niche and spatially limited (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Traditional gravity-fed irrigation systems are constructed to divert meltwater from rivers, streams, or glaciers through unlined or partially lined channels to agricultural terraces. Intakes are often temporary or semi-permanent structures built from stone, earth, or timber, designed to be reconstructed or repaired following seasonal damage. Channels follow natural contours and steep slopes, prioritising reach and coverage over hydraulic efficiency. These systems rely on gravity alone and do not require mechanical pumping or pressurisation (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Design characteristics of traditional systems reflect local environmental conditions and collective labour constraints. Channels are frequently narrow, exposed, and vulnerable to seepage, erosion, and blockage by debris. However, they are also flexible and repairable using locally available materials and skills. Maintenance regimes are embedded in customary institutions, allowing systems to function with minimal external inputs under normal conditions (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Improved or partially modernised systems have emerged through government or donor-supported interventions aimed at reducing losses, improving stability, or extending command areas. These typically involve selective lining of channels, construction of more permanent intakes, and limited use of control structures such as gates or division boxes. While these improvements enhance durability and reduce routine maintenance needs, they remain fundamentally gravity-fed and dependent on upstream hydrological stability (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, Version 1.02, 2024).

Modern irrigation systems, including pressurised drip or sprinkler systems, are rare and confined to small plots, demonstration sites, or kitchen gardens. Field evidence suggests that these systems have been introduced primarily through pilot projects and are not integrated into main irrigation networks. Their adoption is constrained by terrain, lack of pressurised water supply, high capital and maintenance costs, and collective governance arrangements that limit individual incentives

for efficiency gains (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Across all typologies, infrastructure quality is closely linked to exposure to natural hazards. Traditional systems are particularly vulnerable at intake points and along steep channel sections, while improved systems may suffer disproportionate damage when rigid structures fail under extreme events. The literature notes that in mountain environments, design choices that prioritise flexibility and repairability can be as important as structural robustness, particularly where maintenance capacity is limited (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

In summary, irrigation infrastructure in Gilgit-Baltistan is dominated by traditional gravity-fed systems that are well adapted to local conditions but increasingly stressed by climate variability and physical degradation. Improved systems offer incremental gains in durability but do not fundamentally alter system vulnerability. Fully modern irrigation technologies remain marginal and context-specific. This typological understanding provides the foundation for assessing infrastructure quality, reliability, and resilience in subsequent sections.

The principal irrigation system types observed in Gilgit-Baltistan, along with their defining characteristics and vulnerabilities, are summarised in Table 5.

Table 5. Typology of Irrigation Systems in Gilgit-Baltistan

Irrigation System Type	Primary Water Source	Typical Design /Construction Features	Ownership / Management Arrangement	Scale and Coverage	Key Vulnerabilities
Traditional gravity-fed channels	Snowmelt and glacial meltwater	Open channels, often cut through rock or unstable slopes; earthen or stone-lined sections; simple diversion structures	Community-owned and managed under customary institutions	Predominant system across GB, serving small to medium command areas	High exposure to floods, landslides, sedimentation; damage to intakes; labour-intensive maintenance (Field consultations; Annex C).
Improved traditional systems	Snowmelt and glacial meltwater	Traditional alignment with selective lining, improved intakes, or protection works introduced through public or project support	Community-managed, with external support during construction or rehabilitation	Limited but increasing in selected locations	Reduced losses compared to purely traditional systems, but still vulnerable to climate shocks and maintenance constraints (Field consultations; Annex C).
Spring-fed irrigation systems	Springs and local groundwater	Short channels, low-gradient	Community-managed, often at hamlet level	Localised and	Seasonal variability, declining spring discharge, limited

		flows, limited storage		small-scale	expansion potential (Field consultations; Annex C).
Mechanised pumping systems	Rivers or groundwater	Diesel or electric pumps, pipelines, storage tanks	Mixed arrangements; sometimes privately owned or managed by groups	Limited and site-specific	High operating costs, energy dependence, maintenance challenges (Field consultations; Annex C).
Pilot modern irrigation systems (drip/sprinkler)	Pumped or gravity-fed water	Pressurised systems, pipes, emitters or sprinklers	Typically, project-supported and farmer-managed	Very limited and experimental	High capital and maintenance costs; limited suitability for subsistence cropping systems (Field consultations; Annex C; Annex A).

Source: Field consultations and coded summaries, Annex C; regional consultation transcripts; Literature Review, Annex A)

3.2 Technical Standards, Failure Patterns, and Vulnerabilities

Technical standards applied to irrigation infrastructure in Gilgit-Baltistan vary widely and are often shaped more by terrain, material availability, and implementation modalities than by formally codified engineering norms. Most irrigation systems, particularly traditional gravity-fed channels, were constructed through community labour using local materials and indigenous design knowledge, with limited reference to standardised technical specifications. Where public or project-supported interventions have occurred, selective application of engineering standards has been introduced, primarily in the form of lined channels, masonry intakes, or reinforced control structures (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Primary field evidence indicates that the absence of uniform technical standards contributes to recurring patterns of infrastructure failure. Intake structures are the most vulnerable components across system types, frequently damaged or destroyed by high flows, debris, and sediment during peak melt periods and extreme weather events. Temporary or semi-permanent intakes constructed from stone and earth are particularly susceptible, but even more rigid, engineered intakes have failed when design parameters did not adequately account for extreme hydrological variability (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Along conveyance channels, common failure patterns include seepage losses due to unlined or poorly compacted sections, erosion of channel banks on steep slopes, and blockages caused by landslides or debris flows. Field evidence from multiple regions indicates that seepage losses can be substantial, reducing water availability at tail-end command areas and exacerbating inequities

among users. In areas with sandy or unstable soils, channel collapse and frequent breaches are reported, requiring repeated emergency repairs and increasing maintenance burdens on communities (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Improved or partially modernised systems exhibit a different set of vulnerabilities. While lined channels and permanent structures reduce routine maintenance requirements, they can introduce rigidity that increases the scale of damage when failures occur. Field evidence suggests that when lined sections fail, repairs are more complex, time-consuming, and dependent on external technical or financial support. This can result in longer system shutdowns compared to traditional channels that are more easily repaired using local resources (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Technical vulnerabilities are compounded by limited consideration of life-cycle maintenance during design and construction. The literature review highlights that irrigation investments in mountain regions often prioritise initial construction outputs over long-term durability, adaptability, and maintainability. Designs that do not explicitly incorporate ease of repair, sediment management, and flexibility under extreme events tend to perform poorly over time, particularly where institutional support for maintenance is weak (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Climate-related stressors further intensify existing vulnerabilities. Field evidence documents increasing frequency and severity of flood events, glacial melt variability, and landslides affecting irrigation infrastructure. These hazards disproportionately impact upstream intakes and exposed channel sections, leading to repeated system failures within a single irrigation season in some locations. In Diamer and Astore-type contexts, prolonged outages of 20 days or more during critical cropping periods have been reported, resulting in severe yield losses and, in some cases, complete crop failure (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

This review indicates that irrigation infrastructure in Gilgit-Baltistan operates with limited technical redundancy and high exposure to failure. The combination of variable technical standards, predictable failure patterns, and increasing climate stress creates a cycle of damage, repair, and degradation that undermines system reliability. Addressing these vulnerabilities requires not only improved technical design, but also alignment with maintenance capacity, institutional arrangements, and realistic performance expectations. These issues are examined further in the subsequent sections on climate resilience and system integration.

The recurring nature of these technical failures across system types and regions points to identifiable patterns rather than isolated incidents. To synthesise field observations and highlight the most common points of vulnerability, Table 6 summarises the dominant failure modes observed across irrigation infrastructure components, their primary triggers, and their operational consequences. The table draws directly on coded field evidence and consultation transcripts and provides a structured basis for assessing technical risk and prioritising resilience interventions.

Table 6. Common Failure Patterns and Vulnerabilities in Irrigation Infrastructure

Infrastructure Component	Typical Failure Pattern	Primary Triggers / Causes	Operational Consequences	Evidence Source

Intake structures (traditional)	Washed away or blocked during peak flows	High meltwater discharge, debris load, sedimentation	Complete system shutdown until rebuilt	Field consultations (Gilgit, Baltistan, Diamer/Astore); Annex C
Intake structures (engineered)	Structural failure or undermining	Inadequate design for extreme events; rigid construction	Prolonged outages requiring external repair support	Annex C; EU Policy Support TA Inception Report
Unlined conveyance channels	Seepage losses; bank erosion	Sandy soils, steep slopes, poor compaction	Reduced tail-end water availability; inequity	Annex C (multiple regions)
Lined channel sections	Cracking or collapse	Differential settlement; slope instability	Complex repairs; longer downtime	Annex C
Channels in landslide-prone areas	Blockage or burial	Slope failure; intense rainfall events	Emergency maintenance; repeated disruptions	Annex C; field transcripts
Control structures (gates/dividers)	Malfunction or damage	Poor maintenance; debris accumulation	Inefficient water distribution	Field consultations
Entire systems (climate-exposed)	Recurrent seasonal failure	Floods, GLOFs, extreme weather	Crop loss; food security impacts	Annex C; GLOF-related consultations

Source: Field consultations and coded summaries, Annex C; regional consultation transcripts; Literature Review, Annex A.

3.3 Climate Resilience of Existing Infrastructure

The climate resilience of irrigation infrastructure in Gilgit-Baltistan is increasingly tested by changing hydrological regimes, a rising frequency of extreme events, and heightened geomorphological instability. Existing systems—predominantly traditional, gravity-fed channels—were developed to operate under relatively stable seasonal cycles and to be sustained through routine collective repair. While these systems exhibit a degree of adaptive flexibility, their capacity to absorb and recover from repeated or intensified shocks is increasingly constrained.

Primary field evidence indicates that irrigation infrastructure is exposed to multiple, overlapping climate-related hazards, including flash floods, glacial lake outburst floods, accelerated snow and glacier melt, and landslides. These hazards disproportionately affect intake structures and exposed channel sections, resulting in recurrent damage during critical irrigation periods. Communities report that damage events which were historically infrequent now occur multiple times within a single season in some locations, reducing system reliability and substantially increasing maintenance demands (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Seasonal variability in water availability further complicates resilience outcomes. Field evidence suggests that shifts in the timing and intensity of snowmelt have created growing mismatches between water supply and crop water demand. In several areas, high early-season flows cause physical damage to intakes and channels, while late-season reductions in flow affect crop maturation and yields. The absence of storage, regulation, or flow-buffering mechanisms limits the ability of existing systems to manage these intra-seasonal fluctuations (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Improved or partially modernised infrastructure demonstrates mixed resilience performance. While lined channels and permanent intakes reduce routine seepage and minor wear, field evidence indicates that rigid structures can fail catastrophically when exposed to extreme hydrological events or debris flows. Such failures often result in longer service interruptions and higher repair costs compared to traditional systems that rely on more flexible, easily reconstructable elements. This reveals a persistent trade-off between durability under normal operating conditions and adaptability under extreme stress (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

In addition to structural vulnerabilities, a critical but under-addressed constraint in irrigation systems in Gilgit-Baltistan is the absence of water storage and seasonal buffering capacity. Existing systems are predominantly flow-based, relying on real-time diversion of snowmelt and glacial water, which results in excess flows during early melt periods and water deficits during peak crop demand later in the season. Field evidence indicates that this growing mismatch between supply and demand is intensifying under climate variability, leading to inefficient water use, reduced yields, and increased system stress. In the absence of storage, communities depend on rotational allocation and adaptive scheduling, which are increasingly insufficient. Introducing small-scale, decentralized storage options—such as ponds, tanks, and intermediate reservoirs—offers a viable pathway to improve system reliability by smoothing seasonal variability and enhancing resilience, without altering the fundamental design and governance structure of traditional irrigation systems.

The literature review reinforces that climate resilience in mountain irrigation systems depends less on structural robustness alone than on adaptability, redundancy, and rapid repair capacity. Designs that incorporate flexibility, sacrificial components, and ease of reconstruction are often better suited to high-uncertainty environments characterised by frequent disturbance. However, the effectiveness of such adaptive approaches depends on sustained collective action and labour availability—both of which are increasingly constrained by demographic change, outmigration, and repeated shock exposure (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

Institutional factors further shape resilience outcomes. Customary irrigation institutions continue to play a central role in mobilising emergency repairs, but field evidence suggests that the cumulative impact of repeated damage events increasingly exceeds local capacity, particularly when multiple systems are affected simultaneously. Delays in accessing external technical or financial support exacerbate system downtime and agricultural losses. The absence of systematic risk assessment, preventive reinforcement, and contingency financing mechanisms further limits the ability of existing infrastructure to cope with escalating climate risks (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Overall, the climate resilience of irrigation infrastructure in Gilgit-Baltistan is constrained by high physical exposure, design limitations, and institutional capacity gaps. While traditional systems retain important adaptive strengths, the increasing frequency and severity of climate-induced

shocks are eroding their effectiveness. Strengthening resilience therefore requires integrated approaches that embed climate risk considerations into infrastructure design, maintenance planning, and institutional support frameworks. These issues provide a critical foundation for the subsequent analysis of system integration and productivity linkages.

The principal climate-related stressors affecting irrigation infrastructure in Gilgit-Baltistan, together with their exposure pathways and observed system-level impacts, are synthesised in Table 7**.**

Table 7. Climate Stressors, Infrastructure Exposure, and System-Level Impact

Climate Stressor	**Primary Exposure Points in Irrigation Systems**	**Hydrological / Geomorphic Mechanism**	**Observed Infrastructure Impacts**	**System-Level Consequences**	**Evidence Source**
Increased spring and early summer melt flows	River and stream intake structures; headworks	Rapid rise in discharge exceeding design tolerance; high sediment and debris load	Washout or blockage of intakes; diversion failure	Complete system shutdown during early cropping period	Field consultations (Gilgit, Baltistan, Diamer/Astore); Annex C
Intense rainfall events (cloudbursts)	Steep open channels; hillside alignments	Surface runoff concentration; slope saturation	Channel overtopping, bank erosion, localised collapse	Recurrent breaches; emergency repairs; water delivery interruptions	Annex C; regional consultation transcripts
Glacial Lake Outburst Floods (GLOFs)	Upstream intakes; valley-floor conveyance sections	Sudden high-energy flood waves; boulder transport	Destruction of intakes and long channel reaches	Multi-season service disruption; dependence on rehabilitation grants	Annex C; GLOF-related stakeholder consultations
Increased sediment yield	Intakes; settling basins (where present); channels	Enhanced erosion upstream; glacial grinding	Frequent blockage; abrasion of structures	Reduced conveyance efficiency; higher maintenance burden	Annex C; field observations
Landslides and slope failures	Channels cut along steep or unstable slopes	Freeze–thaw cycles; rainfall-triggered instability	Burial or severing of channels	Loss of service to downstream command areas	Annex C; field transcripts

Freeze–thaw cycles	Masonry intakes; lined channel sections	Thermal expansion and contraction	Cracking, joint failure, gradual structural weakening	Progressive deterioration; higher repair complexity	Annex A; field observations
Variability in seasonal water availability	Entire irrigation systems	Shifts in snowmelt timing; inter-annual variability	Mismatch between water supply and crop calendars	Yield losses; reduced system reliability	Annex C; Annex A
Rising temperatures	All exposed infrastructure	Accelerated material degradation; altered melt regimes	Reduced lifespan of structures	Increased long-term O&M requirements	

3.4 Case Studies (AKRSP, GLOF II, ICIMOD)

3.4.1 Purpose and Analytical Framing

The case studies presented in this section illustrate how different institutional and technical approaches to irrigation development and rehabilitation have been applied in Gilgit-Baltistan and comparable high-mountain contexts. The purpose is not to identify best-practice models for replication, but to examine how specific interventions have engaged with traditional irrigation systems, addressed technical vulnerabilities, and interacted with local governance and operation-and-maintenance arrangements.

These cases are selected because they represent distinct intervention logics: long-term community-embedded support (AKRSP), climate-risk-driven infrastructure rehabilitation (GLOF II), and innovation-oriented pilot experimentation (ICIMOD). Together, they provide insight into the strengths and limitations of different approaches when applied in environments characterised by physical fragility, collective water management, and limited institutional capacity.

The cases are analysed with particular attention to (i) alignment with customary institutions, (ii) treatment of operation and maintenance responsibilities, (iii) response to climate-related risks, and (iv) implications for sustainability and scalability. The findings from these cases inform subsequent sections on institutional resilience, financing models, and strategic reform options.

3.4.2 AKRSP-Supported Irrigation Interventions

AKRSP-supported irrigation interventions represent the longest-running and most deeply embedded engagement with community-managed irrigation systems in Gilgit-Baltistan. These interventions typically focused on incremental upgrading of existing traditional systems, including stabilisation of intakes, selective lining of vulnerable channel sections, and extension of command areas. Implementation relied heavily on community labour contributions and decision-making through existing customary arrangements.

Field evidence indicates that this approach strengthened local ownership and ensured that infrastructure design remained compatible with community maintenance capacities. Systems were generally repairable using local skills and materials, allowing for rapid post-damage recovery

under normal conditions. However, the interventions did not fundamentally alter exposure to climate-related hazards, nor did they introduce formal mechanisms for long-term operation and maintenance financing. As climate shocks intensified and labour availability declined due to outmigration, the limitations of relying solely on customary O&M arrangements became more pronounced (Annex C; Annex A).

3.4.3 GLOF II Programme Irrigation Interventions

The GLOF II programme introduced a more explicit climate-risk perspective into irrigation rehabilitation, linking irrigation infrastructure to broader disaster risk reduction objectives. Interventions focused on hazard-prone intakes and channel sections, often incorporating engineered structures designed to withstand high flows, debris, and flood impacts.

Field evidence suggests that these measures reduced immediate vulnerability at targeted sites and improved short-term system stability. However, several limitations were observed. Rigid structures introduced under the programme were often costly to repair when damaged and required external technical support beyond community capacity. In some cases, this resulted in longer service interruptions compared to traditional systems that could be reconstructed quickly using local resources. While the programme successfully elevated climate risk awareness, it highlighted the limits of infrastructure-centric solutions in the absence of strengthened local operation and maintenance institutions and predictable financing mechanisms (Annex C; EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

3.4.4 ICIMOD Pilot Initiatives

ICIMOD-supported pilot initiatives have focused on testing innovative approaches to irrigation design, diagnostics, and climate adaptation, including alternative intake designs, sediment management concepts, and small-scale efficiency technologies. These pilots have contributed valuable technical learning, particularly in understanding system behaviour under variable flow and sediment conditions.

However, field evidence indicates that the scale of these initiatives has remained limited and that uptake beyond pilot sites has been constrained. High capital costs, technical complexity, and limited institutional integration reduced the feasibility of scaling through existing community governance structures. While pilots enhanced technical understanding, their long-term impact depends on stronger linkage with provincial programmes and clearer pathways for embedding innovation within customary management systems (Annex A; field consultations).

3.4.5 Comparative Analysis of Case Studies

The comparative features of the three case study approaches are synthesised in Table 8, highlighting differences in intervention logic, engagement with customary institutions, treatment of operation and maintenance, and implications for long-term sustainability.

Table 8. Comparative Analysis of Selected Irrigation Case Studies

Dimension	AKRSP Interventions	GLOF II Programme	ICIMOD Pilot Initiatives
Primary objective	Agricultural development and livelihoods	Climate risk reduction	Innovation and technical learning

Entry point	Existing community irrigation systems	Hazard-prone infrastructure	Pilot and demonstration sites
Infrastructure approach	Incremental upgrading of traditional systems	Engineered stabilisation and protection	Experimental designs and tools
Engagement with customary institutions	High	Moderate	Variable
Treatment of O&M	Relied on customary practices	Limited institutional strengthening	Not systematically addressed
Technical complexity introduced	Low to moderate	Moderate to high	Often high
Repairability using local resources	High	Often limited	Often limited
Dependency on external support	Low to moderate	Moderate to high	High
Scalability	Gradual and context-dependent	Limited by cost and complexity	Low without institutional embedding
Key strength	Local ownership and continuity	Explicit climate-risk focus	Technical insight and diagnostics
Key limitation	Weak long-term financing and climate buffering	Institutional fragility post-project	Pilot isolation and limited uptake
Evidence source	Annex C; Annex A	Annex C; Inception Report	Annex A; field consultations

3.4.6 Synthesis of Case Study Lessons

Taken together, the case studies demonstrate that irrigation interventions in Gilgit-Baltistan are most effective when they align with existing community governance arrangements and maintenance capacities. Interventions that introduce technical complexity without corresponding institutional adaptation tend to increase dependence on external support and reduce long-term resilience.

The cases also reveal a recurring pattern: infrastructure improvements can temporarily enhance system performance, but sustained resilience depends on the strength, legitimacy, and resourcing of local operation and maintenance institutions. These findings directly inform the subsequent analysis of operations, maintenance, and financing models, where institutional sustainability becomes central.

4. Modern Irrigation Techniques and Efficient Water Use

4.1 Overview of Modern Irrigation Techniques

This section examines the application of selected modern irrigation technologies—solar pumps, drip irrigation, and sprinkler systems—in Gilgit-Baltistan, with particular attention to context specificity, system integration, and co-benefits beyond water-use efficiency. The assessment draws on primary field evidence, project documentation, and relevant literature, and treats observed applications as situational responses rather than generalisable solutions.

Overall, evidence indicates that modern irrigation technologies have not been adopted as standalone system-wide solutions in Gilgit-Baltistan. Where applied, they have functioned as supplementary or adaptive measures, often responding to specific topographic, climatic, or livelihood constraints. Their effectiveness and sustainability depend less on the technologies themselves than on how they are integrated with existing gravity-fed systems, storage arrangements, land use practices, and institutional governance.

4.1.1 Solar Pumps

Solar-powered pumping systems have featured prominently in national and provincial policy discourse as a means of expanding irrigation access in off-grid and remote areas. In Gilgit-Baltistan, however, field evidence shows that their application has been limited and highly context-dependent. Most observed installations serve small plots, kitchen gardens, or supplementary irrigation needs, rather than forming part of main irrigation networks.

Importantly, evidence suggests that solar pumps are rarely used to access groundwater. Instead, where they have demonstrated potential, they are employed to lift surface water—drawn from rivers, streams, or existing irrigation channels—over short vertical distances. In several locations, solar pumps have been used to raise water to elevated storage tanks, from which water is then distributed by gravity. This configuration reduces the need for continuous pumping, lowers operational complexity, and aligns more closely with existing gravity-based irrigation logic.

Such arrangements offer multiple co-benefits. Stored water enables more reliable supply during periods of channel disruption or diurnal flow variability, supports controlled application through drip or sprinkler systems, and facilitates irrigation of terraced slopes and marginal lands. In some cases, gravity-fed distribution from elevated tanks has been used to support tree planting and grass establishment on unstable slopes, contributing to erosion control, slope stabilisation, and microclimate regulation alongside irrigation.

Despite this potential, several constraints limit wider adoption. Solar pumping introduces new operation and maintenance requirements—panel cleaning, pump servicing, and component replacement—that are not well aligned with customary labour-based maintenance systems. Field evidence indicates that once external support ends, communities often face difficulties sustaining these systems. In addition, the absence of clear technical guidance and institutional frameworks for integrating solar pumping into collective irrigation governance creates uncertainty around ownership, cost-sharing, and responsibility.

Overall, solar pumps show promise as enabling technologies within hybrid gravity-based systems, particularly when used for surface water lifting and storage. However, their application remains experimental, and further research, design adaptation, and piloting are required to assess long-term performance, institutional fit, and scalability.

4.1.2 Drip Irrigation

Drip irrigation systems have been introduced in Gilgit-Baltistan primarily through pilot and demonstration projects supported by government programmes and development partners. Field evidence shows that these systems are typically confined to small plots, horticultural gardens, or household-level cultivation, often disconnected from main irrigation channels.

Technical constraints remain significant. Drip systems require relatively clean and stable water supply, filtration, and controlled pressure—conditions that are rarely met in sediment-laden, gravity-fed irrigation networks. Seasonal flow variability and frequent disruptions caused by channel damage further undermine system reliability. Where drip systems depend on temporary storage or manual filling, labour requirements increase and performance becomes inconsistent.

Economic and institutional factors further limit adoption. Drip irrigation involves high upfront capital costs and recurring maintenance expenses, making it economically viable mainly for high-value crops. In Gilgit-Baltistan, agriculture remains largely subsistence-oriented, with staple crops dominating production systems. The availability of subsidised wheat further weakens incentives for investment in capital-intensive irrigation technologies to achieve marginal efficiency gains.

In community-managed irrigation systems, individual investment in drip irrigation does not necessarily translate into private water savings. Water conserved at one plot is typically redistributed downstream under customary allocation rules, reducing incentives for individual adoption. Evidence suggests that without collective agreement and system-wide coordination, drip irrigation is unlikely to deliver aggregate efficiency gains.

Nevertheless, where drip systems are integrated with small storage tanks supplied by gravity-fed or solar-lifted water, they have shown potential to support horticulture, kitchen gardening, and tree establishment in water-scarce or topographically constrained locations. These applications point to the need for further piloting of collective or semi-collective drip systems linked to storage and land management objectives rather than plot-level efficiency alone.

4.1.3 Sprinkler Systems

Sprinkler irrigation systems have seen limited application in Gilgit-Baltistan, primarily through short-term demonstrations and small-scale trials. Field evidence indicates that their performance is constrained by uneven terrain, terraced fields, wind exposure, and pressure requirements. Wind conditions common in mountain valleys reduce application efficiency, while topographic variation complicates uniform water distribution.

Sprinkler systems also depend on pressurised water supply, which is largely absent in existing irrigation networks. Where pressure is generated through pumps, additional energy and maintenance requirements arise, creating dependencies that are difficult to sustain under community management. As with drip systems, sprinkler irrigation has not been integrated into collective irrigation planning or governance arrangements, limiting its relevance beyond isolated applications.

The literature suggests that sprinkler systems are generally better suited to relatively flat terrain and controlled farming environments. In high-mountain contexts, their applicability appears limited to niche uses, such as supplementary irrigation of fodder plots or tree plantations where gravity-fed distribution from elevated storage is feasible (Figure).

Figure 5. Solar Lift-Storage-Gravity Systems



4.1.4 Overall Assessment and Research Needs

Across solar pumps, drip irrigation, and sprinkler systems, evidence indicates that application in Gilgit-Baltistan remains fragmented, experimental, and highly context-specific. These technologies have not been adopted at scale, nor have they fundamentally altered system-level water-use efficiency. The primary constraints are not lack of awareness or technical knowledge, but structural factors, including unreliable water delivery, subsistence-oriented production systems, collective governance arrangements, and weak economic incentives.

At the same time, field experience points to emerging hybrid configurations—particularly the use of solar pumps for surface water lifting, storage in elevated tanks, and gravity-based distribution through drip or sprinkler systems—that warrant further research and piloting. Such configurations offer co-benefits for water reliability, slope stabilisation, tree and grass establishment, and climate adaptation, extending the value of irrigation beyond crop productivity alone (Figure 4).

Future innovation efforts should therefore focus less on promoting individual technologies in isolation and more on system-level design, institutional integration, and adaptive piloting tailored to local hydrological and governance contexts. Without such an approach, modern irrigation technologies are likely to remain confined to pilots and demonstrations rather than contributing meaningfully to irrigation resilience and sustainability in Gilgit-Baltistan.

4.3 Digital Tools: GIS, Flow Monitoring, Automation

Digital tools have been introduced in the irrigation and water management space in Gilgit-Baltistan primarily through externally supported projects and pilots, with objectives ranging from improved planning and diagnostics to enhanced monitoring and early warning. These tools include geographic information systems (GIS) for mapping irrigation networks and command areas, basic flow monitoring instruments, and limited experimentation with automation or sensor-based technologies. Their application, however, remains uneven and largely disconnected from routine irrigation management.

Primary field evidence indicates that GIS tools have been used mainly for baseline assessments, project planning, and documentation of irrigation assets. Mapping exercises have helped visualise command areas, intake locations, and exposure to hazards, particularly in the context of climate risk and disaster response initiatives. While these tools have improved the quality of diagnostic information available to project teams and planners, their use has rarely translated into sustained

operational decision-making at the community or departmental level once project activities conclude (Annex C: Summary of Primary Evidence from the Field).

Flow monitoring has been introduced in a limited and largely experimental manner. Simple gauges or measurement structures have been installed at selected intakes or channel sections to observe seasonal variability and peak flows. Field evidence suggests that these instruments are not systematically maintained or integrated into allocation or scheduling decisions. In community-managed systems governed by customary water-sharing arrangements, water distribution continues to rely on established rotational rules and social enforcement rather than quantified flow data. As a result, monitoring information, where available, remains informational rather than operational (Annex C: Summary of Primary Evidence from the Field).

Automation and sensor-based technologies are at a very early stage of application in Gilgit-Baltistan. Pilots involving automated gates, remote sensors, or telemetry have been limited in scope and duration. Field evidence indicates that such technologies face significant constraints related to power supply, harsh environmental conditions, technical complexity, and maintenance requirements. Communities and local institutions generally lack the capacity and resources to operate and repair automated systems without continuous external support, leading to rapid degradation once pilot phases end (Annex C: Summary of Primary Evidence from the Field).

The literature review highlights that digital tools can enhance irrigation efficiency and resilience when embedded within supportive institutional frameworks and linked to decision-making processes. In contexts where governance structures, incentives, and capacities are weak, digitalisation risks becoming an add-on rather than a transformative input. Studies caution against technology-driven approaches that prioritise data generation without clear pathways for use, particularly in small-scale, community-managed irrigation systems (Annex A: Literature Review).

Institutional and incentive-related constraints are central to understanding the limited impact of digital tools in Gilgit-Baltistan. Provincial departments face capacity gaps in data management, analysis, and routine use of digital platforms, while community institutions operate through norms and practices that do not depend on quantified data. Without explicit mechanisms to integrate digital outputs into planning, maintenance prioritisation, or resource allocation, the practical value of these tools remains limited (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Overall, digital tools such as GIS, flow monitoring, and automation offer potential benefits for planning, risk assessment, and system understanding, but their current application in Gilgit-Baltistan has not resulted in sustained improvements in irrigation efficiency or management. Their effectiveness is constrained less by technology availability than by institutional readiness, maintenance capacity, and absence of demand for data-driven decision-making. Strengthening these enabling conditions is therefore a prerequisite for any meaningful expansion of digital tools in the irrigation sector, and links directly to the discussion on capacity building and awareness in the following section.

4.4 Barriers and Enablers for Adoption

The adoption of modern irrigation technologies and digital tools in Gilgit-Baltistan is shaped by a combination of technical, economic, institutional, and behavioural factors. Evidence from field consultations and project experience indicates that low uptake is not primarily the result of lack of awareness or resistance to change, but rather reflects structural conditions that constrain both individual and collective incentives to adopt new practices.

A central barrier relates to the collective nature of irrigation systems. Irrigation in Gilgit-Baltistan is predominantly managed through communal arrangements, where water is shared according to established rotational or proportional rules. In such systems, individual farmers have limited incentive to invest in efficiency-enhancing technologies unless adoption is coordinated across the command area. Field evidence indicates that when only a subset of farmers adopts water-saving practices, the water conserved is redistributed downstream rather than retained, resulting in no tangible private benefit for early adopters. This weakens incentives for individual investment and limits spontaneous diffusion of modern technologies (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Economic barriers are also significant. Modern irrigation technologies, including drip and sprinkler systems, involve high upfront capital costs and ongoing maintenance expenditures. Field evidence shows that such investments are only economically viable where farmers produce high-value or market-oriented crops. In Gilgit-Baltistan, agriculture remains largely subsistence-based, with limited diversification and weak market integration. The widespread availability of subsidised wheat as the primary staple further reduces incentives to intensify production or invest in irrigation and farm technologies aimed at increasing water productivity (National Water Policy of Pakistan, 2018; National Water Conservation Strategy of Pakistan, 2023–2027).

Technical and infrastructural constraints further limit adoption. Existing irrigation systems are gravity-fed, unpressurised, and subject to frequent disruption from floods and landslides. Modern irrigation technologies and digital tools typically require reliable water delivery, stable pressure, and regular maintenance. Field evidence indicates that these prerequisites are rarely met, making sustained operation difficult even where initial installation is successful. Harsh environmental conditions also accelerate wear and tear of equipment, increasing maintenance burdens beyond local capacity (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Institutional barriers play a critical role. Customary irrigation institutions are effective in managing water distribution and mobilising labour for routine maintenance, but they are not designed to manage capital-intensive technologies, digital systems, or complex maintenance regimes. At the same time, provincial departments face capacity constraints in providing sustained technical support, extension services, and follow-up. The absence of formal mechanisms linking technology adoption with institutional support and financing results in pilot-driven interventions that are difficult to sustain beyond project timelines (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Despite these barriers, several enabling factors can be identified. Strong social cohesion and established collective action mechanisms provide a foundation for coordinated adoption where incentives are aligned at system level. Field evidence suggests that communities are willing to adopt new practices when benefits are clearly shared, risks are manageable, and support is reliable. Targeted demonstrations linked to collective decision-making processes, rather than individual plots, have shown greater acceptance (Annex C: Summary of Primary Evidence from the Field, EU Policy Support TA – Irrigation Sector Study, 2025).

Policy frameworks at national level also provide enabling signals by emphasising water conservation, climate resilience, and efficiency. However, the literature highlights that these frameworks need to be translated into locally appropriate instruments, financing mechanisms, and institutional arrangements to become effective enablers. Capacity building efforts that focus not only on technical skills but also on governance, coordination, and incentive design are more

likely to support sustained adoption (Annex A: Literature Review, EU Policy Support TA – Irrigation Sector Study, 2025).

In summary, adoption of modern irrigation technologies and digital tools in Gilgit-Baltistan is constrained by collective action dynamics, economic incentives, infrastructural limitations, and institutional capacity gaps. Enabling adoption requires moving beyond technology promotion to address system-level incentives, coordinated decision-making, and sustained support mechanisms. These findings underscore the need for realistic, context-sensitive approaches to irrigation efficiency and inform subsequent discussion on operations, maintenance, and performance frameworks. The principal barriers and enabling conditions influencing the adoption of modern irrigation technologies and digital tools in Gilgit-Baltistan are synthesised in Table 9.

Table 9. Barriers and Enablers to Adoption of Modern Irrigation Technologies

Dimension	Key Barrier Identified	Underlying Mechanism	Enabling Condition (Observed or Potential)	Evidence Source
Institutional (collective systems)	Weak individual incentives	Water saved by one user redistributed within communal systems	System-level or command-area adoption mechanisms	Annex C; Annex A
Economic	High capital and maintenance costs	Subsistence cropping; limited cash flow	Linkage to high-value crops and markets	Annex C
Policy incentives	Low motivation to intensify production	Availability of subsidised wheat	Targeted incentives tied to efficiency gains	National Water Policy 2018; NWCS 2023–27
Technical	Unreliable water delivery	Frequent channel damage; variable flows	Upstream infrastructure stabilisation	Annex C
Energy & maintenance	Dependence on pumps and components	Limited local repair capacity	Technologies aligned with local O&M capacity	Annex C
Institutional capacity	Customary bodies not designed for capital assets	Informal governance; no formal revenue systems	Institutional upgrading and technical backstopping	Inception Report; Annex A
Digital readiness	Limited use of data in decision-making	Governance based on norms, not metrics	Digital tools as support, not substitutes	Annex A
Social factors	Risk aversion	High exposure to failure and loss	Collective demonstrations with shared benefits	Annex C

4.5 Selecting and Adapting Irrigation Innovations in High-Mountain Contexts

The applicability of irrigation technologies in Gilgit-Baltistan depends not only on the type of technology, but on how well it can be adapted to the physical, hydrological, and institutional conditions of high-mountain environments, particularly under increasing climate-induced risks. Primary field evidence and comparative experience indicate that technologies designed for stable, lowland irrigation systems perform poorly when transferred without modification to contexts

characterised by steep terrain, unstable geology, high sediment loads, and extreme flow variability.

In such environments, innovation is best understood as the adaptation and combination of technologies rather than their direct substitution for traditional systems. Gravity-fed irrigation channels remain the backbone of agricultural production and food security in Gilgit-Baltistan and cannot be replaced at scale. However, climate stressors—including flash floods, glacial lake outburst floods, landslides, and erratic seasonal flows—are increasing the vulnerability of long open channels and intakes. This necessitates the selective adoption of supplementary technologies and design adaptations that reduce exposure, diversify water access, and improve resilience.

Field evidence suggests that the most viable innovations are those that (i) minimise dependence on long, exposed conveyance structures; (ii) allow modular and incremental investment; (iii) align with local operation and maintenance capacities; and (iv) generate additional landscape-level benefits such as slope stabilisation, vegetation cover, and flood attenuation.

In this context, solar-powered water lifting combined with local storage and gravity-based micro-distribution emerges as a particularly relevant supplementary approach, alongside improved gravity systems and selective use of piped conveyance in hazard-prone reaches.

Equally important are design choices. Intake structures, abstraction points, and conveyance systems must be designed to tolerate failure, allow rapid repair, and accommodate extreme events rather than resist them rigidly. Technologies that require stable pressure, clean water, continuous energy supply, or complex maintenance regimes are less suitable unless adapted substantially or supported by strong institutional and financial arrangements.

Overall, irrigation innovation in Gilgit-Baltistan is most viable where technologies are selected and adapted to reduce exposure to climate-induced risks while remaining compatible with local governance and maintenance capacities, rather than being introduced as stand-alone technical solutions.

Table 10 below synthesises how different irrigation technologies and hybrid systems can be adopted or adapted in Gilgit-Baltistan in relation to the climate-related vulnerabilities identified earlier, and identifies the conditions under which they are technically and institutionally feasible.

Table 10. Adaptability of Irrigation Technologies in High-Mountain Contexts

Technology / System	Potential Application in Gilgit-Baltistan	Climate-Related Vulnerabilities Addressed	Residual or New Risks	Key Design Adaptations Required	Institutional Fit	Feasibility Assessment
Improved traditional gravity-fed channels	Core subsistence and mixed farming systems	Seasonal variability; minor sediment loads	High exposure at intakes; landslides	Flexible or sacrificial intakes; segmented channels; bypass sections	Strong (customary institutions)	High (with adaptive design)
Solar lift–storage–	Slopes, riverbeds,	Reduces reliance on vulnerable long	Pump failure; abstraction impacts	Debris-protected abstraction; overhead	Moderate to strong (small	High (as a supplement)

gravity systems	terraces above channels	channels; diversifies access		storage; gravity distribution	groups / utilities)	
Gravity-fed piped conveyance	Landslide- or flood-prone reaches	Channel erosion; seepage; localised failures	Pipe rupture in mass movement	Buried pipes; flexible joints; anchor blocks; inspection chambers	Strong	Moderate to high
Gravity-fed drip irrigation	Horticulture, agroforestry, fodder plots	Water scarcity; late-season shortages	Clogging; supply interruptions	Filtration; buffer tanks; low-pressure design	Moderate	Moderate (niche use)
Micro-sprinklers (low pressure)	Orchards, fodder on slopes	Inefficient surface irrigation	Wind losses; mechanical damage	Wind protection; zoned application	Moderate	Moderate (site-specific)
Pressurised sprinkler systems	Field crops	None of the major identified vulnerabilities	Energy dependence; fragility	Stable pressure; energy backup	Weak	Low
Solar pumps (direct application, no storage)	Individual plots only	Limited access where gravity unavailable	Over-abstraction; inequity; O&M burden	Regulation; collective abstraction rules	Weak	Low
Digital tools (GIS, flow monitoring)	Planning, diagnostics, risk assessment	Poor visibility of hazards; reactive response	Data under-use	Simple instruments; institutional integration	Moderate (government-led)	Moderate (supporting role)

5. Operations, Maintenance, and Financing Models

5.1 Traditional O&M Practices and Community Norms

Operation and maintenance of irrigation systems in Gilgit-Baltistan are predominantly governed through customary arrangements rooted in collective responsibility, social obligation, and locally enforced norms. Across regions, routine operation of channels, intakes, and distribution structures is organised through community-agreed schedules that regulate water turns, seasonal allocations, and maintenance responsibilities. These arrangements are typically informal but well understood within communities, relying on elders, water masters, or committee members to coordinate activities and resolve disputes (Annex C: Summary of Primary Evidence from the Field; Ostrom, Governing the Commons, 1990; Kreutzmann, Irrigation and Water Management in the Karakoram, 2011).

Maintenance practices are largely labour-based and episodic, concentrated around critical agricultural seasons or following damage caused by floods, landslides, or channel breaches. Communities mobilise collective labour (hashar or equivalent practices) to clear silt, repair embankments, and restore flow, with participation enforced through social sanctions rather than formal contracts. Primary field evidence indicates that these arrangements have historically been effective in sustaining systems under conditions of low cash availability and strong social cohesion, particularly in small command areas with stable user groups (Annex C; IUCN Pakistan, Customary Water Management Practices in Northern Pakistan, 2000).

Field evidence also highlights increasing stress on traditional O&M systems. Respondents across regions reported that more frequent climate-induced damage, ageing infrastructure, and outmigration—particularly of younger male labour—have reduced the reliability of voluntary maintenance contributions. In several locations, delays in repairs, uneven participation, and growing dependence on external assistance for major rehabilitation works were observed. These pressures are more pronounced in systems serving larger command areas or multiple settlements, where coordination is more complex and social enforcement mechanisms are weaker (Annex C; ICIMOD, Water Management and Climate Risks in the HKH Region, 2019).

Customary norms governing O&M are closely linked to local concepts of equity and collective entitlement rather than efficiency or cost recovery. Maintenance obligations are generally uniform across households, regardless of landholding size or water use intensity, and financial contributions remain rare or symbolic. While this reinforces social cohesion and social legitimacy, it also constrains the ability of communities to mobilise cash resources for preventive maintenance, technical upgrades, or professional services. Comparable patterns are documented across communal irrigation systems in mountain and subsistence-oriented contexts, where labour-based maintenance represents a rational institutional adaptation but becomes increasingly fragile as technical and financial demands rise (Ostrom, Governing the Commons, 1990; Pradhan, Institutions, Incentives and Irrigation, 2010).

Overall, traditional O&M practices in Gilgit-Baltistan continue to form the backbone of irrigation system functioning and provide a strong foundation of local legitimacy and collective action. At the same time, both primary field evidence and comparative documentary sources indicate that these practices alone are increasingly insufficient to address rising maintenance needs, financing gaps, and climate-related risks. This tension between social sustainability and technical adequacy provides the analytical context for examining the role of institutional support, cost-sharing mechanisms, and emerging financing models in the following sections (Annex C; World Bank, Sustaining Irrigation Services: Institutional and Financing Reforms, 2015).

5.2 Informal Community Systems and External Support Mechanisms

Primary documents reviewed and field consultations indicate that irrigation operation and maintenance in Gilgit-Baltistan are governed through well-established customary institutions whose terminology and organisational forms vary by region, but whose functions are broadly comparable. In Gilgit and the Hunza–Nagar region, respondents consistently identified the mirab (also referred to locally as chhumik or water master) as the central figure responsible for regulating water allocation, enforcing rotational rules, and mediating day-to-day disputes. This role derives authority from community consensus and long-standing practice rather than statutory appointment and remains central to irrigation management across valleys in this region (Annex C: Gilgit & Hunza–Nagar; IUCN Pakistan, Customary Water Management Practices in Northern Pakistan, 2000).

In Baltistan, similar functions are performed by locally recognised water managers—often referred to as mirab or region-specific variants—working in close coordination with village elders or councils. Field consultations indicate that these arrangements are embedded in long-standing social and, in some locations, lineage-based norms, with collective decision-making forums convened to address seasonal scheduling, major repairs, and enforcement issues. Collective labour mobilisation, through locally organised equivalents of the hashar system, remains the principal mechanism for routine maintenance and emergency response (Annex C: Baltistan; Kreutzmann, Irrigation and Water Management in the Karakoram, 2011).

In Diamer and Astore, respondents described irrigation management as being overseen by locally recognised canal custodians or water overseers, although the term mirab was used less consistently. Here, authority over water allocation and maintenance is more closely intertwined with tribal or clan-based governance structures, with elders' councils playing a central role in enforcing participation and resolving disputes. Collective labour mobilisation for channel maintenance and repairs is typically organised along kinship or settlement lines rather than through standing village-wide bodies (Annex C: Diamer & Astore; EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Across all regions, Local Government institutions, including the Local Government and Rural Development Department (LG&RD), were rarely identified by respondents as playing a role in routine irrigation management. Where mentioned, their involvement was limited to facilitation of small-scale development schemes or administrative coordination and did not extend to regular operation, maintenance, or enforcement of irrigation rules. Neither field consultations nor primary documents reviewed indicate that Local Government entities hold a formal mandate, technical capacity, or recurrent financing arrangements for irrigation O&M (Annex C; EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Semi organised community structures such as Village User Associations, established under development programmes, are present in some locations but remain embedded within broader customary systems. These bodies function as semi-formal extensions of informal governance arrangements and derive their authority primarily from social legitimacy rather than legal mandate. Field evidence suggests that their role is more pronounced in interfacing with external agencies and managing project-related activities, while responsibility for routine operation and maintenance continues to rest with region-specific customary institutions such as the mirab, elders' councils, and clan-based governance structures (Annex C; AKRSP, Northern Areas Development Experience, 2002).

Taken together, the evidence demonstrates that irrigation sustainability in Gilgit-Baltistan is anchored in named, regionally differentiated customary institutions rather than in uniform formal administrative structures. Recognising these institutional variations is essential for any effort to strengthen operation and maintenance arrangements in ways that are both effective and locally legitimate.

To assess the relative prevalence of different operation and maintenance arrangements, the primary consultation evidence was reviewed and coded across regions. Table 11 summarises the dominant O&M modalities observed in Gilgit, Baltistan, and Diamer/Astore.

Table 11. Dominant O&M Arrangements by Region

O&M Arrangement	Gilgit	Baltistan	Diamer / Astore
O&M led primarily by customary institutions (elders, mirab, water committees)	~90%	~95%	~85%
Routine O&M carried out through collective labour (hashar or equivalent)	~85%	~90%	~80%
Presence of formally written O&M rules or by-laws	<20%	<10%	<15%
Routine involvement of public departments in day-to-day O&M	<5%	<5%	<5%
Regular public financing for O&M (non-project, non-emergency)	<5%	<5%	<5%

Source: Primary field consultations synthesised in Annex C (C.1–C.3) and regional transcripts.

The table confirms that irrigation O&M across Gilgit-Baltistan remains overwhelmingly anchored in customary institutions and collective labour, with formal rules, routine public involvement, and recurrent financing playing a negligible role.

5.3 Emerging Cost-Sharing and Financing Models

Primary documents reviewed and field consultations indicate that formal cost-sharing and financing mechanisms for irrigation operation and maintenance remain limited and uneven across Gilgit-Baltistan. In most locations, routine O&M continues to be financed almost entirely through unpaid collective labour rather than through monetary contributions or structured user fees. Cash-based financing is generally mobilised only in exceptional circumstances, such as major repairs following flood damage or partial system reconstruction, and even then, contributions are typically ad hoc and negotiated case by case (Annex C: Summary of Primary Evidence from the Field; IUCN Pakistan, Customary Water Management Practices in Northern Pakistan, 2000).

Field consultations revealed a small number of emerging practices where communities have begun to supplement labour contributions with modest financial inputs. These include occasional household cash contributions for the purchase of construction materials, hiring of skilled labour for technically complex repairs, or co-financing arrangements linked to externally supported rehabilitation projects. Such practices are not standardised and remain highly dependent on local leadership, social cohesion, and the perceived urgency of repairs. Respondents emphasised that willingness to contribute cash is closely tied to trust in local decision-makers and confidence that funds will be used transparently (Annex C).

To assess how far communities have moved beyond labour-based maintenance toward monetary cost-sharing, consultation responses were reviewed for evidence of existing financing practices.

Table 12 summarises the prevalence of different cost-sharing and financing arrangements observed across regions.

Table 12. Prevalence of Cost-Sharing and Financing Practices (By region)

Financing / Cost-Sharing Practice	Gilgit	Baltistan	Diamer / Astore
Routine O&M financed primarily through unpaid collective labour	~90%	~95%	~85%
Household cash contributions for routine O&M	<10%	<5%	<10%
Cash contributions mobilised only during major repairs or flood damage	~35%	~40%	~45%
Community purchase of materials using pooled cash	~20%	~25%	~20%
Hiring of skilled labour using community funds	~15%	~20%	~15%
Co-financing linked to externally supported projects (NGOs / donors)	~25%	~30%	~20%
Existence of recurrent user fees or formal tariffs	<5%	<5%	<5%
Dedicated Local Government budget for routine irrigation O&M	0%	0%	0%

Source: Primary field consultations synthesised in Annex C (C.1–C.3); regional transcripts; EU TA Inception Report.

In areas with prior engagement by development programmes, including projects supported by the Aga Khan Rural Support Programme and other NGOs, limited experimentation with cost-sharing arrangements has been observed. These arrangements have typically involved partial community contributions—either in cash or labour—complementing external grants or technical assistance. However, primary documents and field evidence indicate that these models have not evolved into recurrent financing mechanisms for routine O&M. Once external project support ends, communities generally revert to customary labour-based maintenance practices rather than sustaining monetary contributions (AKRSP, Northern Areas Development Experience, 2002; World Bank, Sustaining Irrigation Services: Institutional and Financing Reforms, 2015).

The role of public financing in irrigation O&M at the local level is limited. Field consultations did not identify systematic budgetary allocations from Local Government institutions or the Local Bodies and Rural Development Department for irrigation maintenance. Where public funds were mentioned, they were associated with discrete development schemes or emergency interventions rather than with predictable or recurrent O&M financing. There is no evidence from primary documents that irrigation systems in Gilgit-Baltistan are supported through dedicated local revenue streams or formal cost-recovery mechanisms (Annex C; EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Table 13. Frequency of Reported Constraints to Monetary Cost-Sharing (By region)

Reported Constraint	Gilgit	Baltistan	Diamer / Astore
Preference for labour contributions over cash	~85%	~90%	~80%
Limited household cash availability	~70%	~75%	~80%
Low trust in fund management or fear of misuse	~40%	~45%	~50%
Absence of formal or agreed cost-sharing rules	~80%	~85%	~75%
Perception that O&M is a collective obligation, not a paid service	~75%	~80%	~70%
Irregular and unpredictable maintenance needs (shock-driven)	~60%	~65%	~70%
Lack of linkage to public or project co-financing	~55%	~60%	~65%
Weak engagement of Local Government in O&M financing	~90%	~95%	~90%

Source: Primary field consultations synthesised in Annex C (C.1–C.3); regional transcripts.

Comparative literature reviewed under this study suggests that such reliance on labour-based maintenance and episodic cost-sharing is common in subsistence-oriented, communal irrigation systems, particularly in mountain regions. While these arrangements minimise financial burdens on households, they also constrain the capacity of systems to address growing maintenance demands, invest in preventive works, or respond effectively to climate-related risks. Evidence from other regions indicates that transitions toward more sustainable financing typically require gradual, locally negotiated approaches that build on existing norms rather than impose uniform fee structures (Ostrom, Governing the Commons, 1990; IFAD, Sustainable Smallholder Irrigation Development, 2016).

Overall, emerging cost-sharing and financing practices in Gilgit-Baltistan remain fragmented, informal, and highly context-specific. While there are early indications of willingness among some communities to engage in limited monetary contributions under specific conditions, these practices have not yet coalesced into stable or scalable financing models. Strengthening irrigation sustainability will therefore require careful exploration of hybrid approaches that combine customary labour systems with modest, transparent financial mechanisms, aligned with local capacities and institutional arrangements.

Where user contributions are agreed locally, digital payment channels can be used selectively to improve transparency, record-keeping, and convenience in the collection of service charges, without altering customary decision-making or contribution norms.

5.4 Public Sector Support and Partnership Approaches

Primary documents reviewed and field consultations indicate that public sector institutions play a limited role in the routine operation and maintenance of community-owned irrigation systems in Gilgit-Baltistan. Day-to-day management, seasonal maintenance, and emergency repairs are overwhelmingly undertaken through customary community arrangements, with little direct

involvement by line departments or local government bodies (Annex C: Summary of Primary Evidence from the Field).

Where public sector engagement is evident, it is largely confined to episodic and targeted support, most commonly in the form of small grants or rehabilitation assistance for infrastructure damaged by floods, glacial lake outburst floods (GLOFs), landslides, or other natural disasters. Field consultations identified instances where Local Government and Local Bodies & Rural Development Department resources were mobilised to support reconstruction or partial rehabilitation of damaged channels and intakes. However, such support is typically project-based, limited in scale, and does not extend to ongoing O&M responsibilities or recurrent financing (Annex C; EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Despite this limited operational role, the evidence underscores the critical importance of public sector involvement in enabling robust and sustainable institutional systems for irrigation management. Primary documents reviewed emphasise that community-led systems require an enabling environment that provides legal recognition, clarity of roles, access to technical guidance, and linkages to financing and disaster response mechanisms. Without such support, informal systems—while resilient—remain vulnerable to increasing climate risks, infrastructure ageing, and coordination failures beyond the capacity of communities to manage alone (World Bank, Sustaining Irrigation Services: Institutional and Financing Reforms, 2015; IFAD, Sustainable Smallholder Irrigation Development, 2016).

Field consultations further suggest that communities do not expect the public sector to assume responsibility for operating irrigation systems, but they do recognise the value of public institutions in setting rules, supporting institutional strengthening, and facilitating access to resources during periods of stress. This distinction between ownership and operation by communities and institutional stewardship by the public sector is central to understanding feasible partnership models in the GB context (Annex C).

Overall, the evidence indicates that future partnership approaches should not be framed primarily in terms of outsourcing O&M or introducing conventional PPP models. Instead, the role of the public sector lies in designing and supporting institutional frameworks that strengthen community governance, enable cost-sharing mechanisms, improve technical resilience, and integrate irrigation systems into broader water management and disaster risk reduction strategies.

5.5 Lessons from Regional and Global Experience for Sustainable O&M

Regional and global experience with small-scale, community-managed irrigation systems offers several relevant lessons for strengthening operation and maintenance in contexts comparable to Gilgit-Baltistan. Evidence from mountain and resource-constrained environments in South Asia, Central Asia, and parts of Latin America consistently shows that the sustainability of communal irrigation systems depends less on initial infrastructure quality than on the robustness of locally embedded O&M arrangements and the institutional environment that supports them. These lessons are particularly pertinent for settings where irrigation infrastructure is predominantly community-owned, agriculture remains subsistence-oriented, and formal service delivery institutions have limited reach.

A first recurring lesson is that customary O&M systems, while resilient, have definable limits. Across regions, collective labour-based maintenance and socially enforced compliance have proven effective where systems are small, social cohesion is strong, and technical demands are modest. However, comparative evidence shows that such arrangements become increasingly

strained as infrastructure ages, climate-related damage intensifies, and labour availability declines due to livelihood diversification and migration. Similar to the situation observed in Gilgit-Baltistan, these pressures often lead to deferred maintenance, uneven participation, and growing dependence on external assistance for major repairs (Kreutzmann, Irrigation and Water Management in the Karakoram, 2011; IUCN Pakistan, Customary Water Management Practices in Northern Pakistan, 2000).

A second lesson is that sustainable O&M rarely emerges from abrupt institutional replacement. Experiences from Central Asia and Latin America demonstrate that attempts to impose fully formalised management structures or cost-recovery mechanisms on communal systems frequently undermine local legitimacy and weaken compliance. In contrast, more successful cases involved gradual, incremental upgrading of existing customary arrangements. These transitions often retained traditional roles and norms while introducing limited formalisation, such as clearer rules for contributions, basic financial management, or access to technical services. This reinforces the importance of working with, rather than against, established community institutions (Ostrom, Governing the Commons, 1990; World Bank, Sustaining Irrigation Services: Institutional and Financing Reforms, 2015).

A third lesson concerns the relationship between financing and governance. Comparative evidence indicates that sustainable financing for O&M in communal irrigation systems is rarely achieved through uniform user fees or market-based tariffs alone, particularly in subsistence-oriented economies. Instead, hybrid arrangements—combining labour contributions, modest cash inputs, and targeted public support—have been more durable. Crucially, the willingness of users to contribute financially is closely linked to trust in governance arrangements, transparency in decision-making, and perceived fairness in water distribution. Where these conditions are absent, even technically sound financing models have failed to take root (IFAD, Sustainable Smallholder Irrigation Development, 2016).

Finally, regional experience highlights the importance of public sector stewardship rather than direct operation. In comparable contexts, the most effective public interventions focused on enabling functions: providing legal recognition to community entities, setting basic technical and safety standards, supporting institutional learning, and facilitating access to financing and disaster response mechanisms. Direct public operation or outsourcing of O&M to private actors has generally been infeasible in low-revenue, high-risk environments. This lesson aligns closely with the GB context, where communities retain ownership and responsibility for irrigation systems but require an enabling institutional framework to adapt to growing technical and climatic challenges.

Taken together, regional and global experience reinforces the conclusion that strengthening O&M in Gilgit-Baltistan will require incremental, institutionally grounded approaches rather than wholesale adoption of external models. Sustainable outcomes are most likely where customary systems are explicitly recognised, selectively upgraded, and supported through coherent public stewardship, rather than displaced or bypassed. These lessons provide a comparative lens for interpreting the findings of this study and inform the institutional pathways discussed in the following subsection.

5.6 Institutionalization and Professionalization of Operation and Maintenance

Evidence from Gilgit-Baltistan and comparable contexts indicates that strengthening the sustainability of community-managed irrigation systems requires institutionalisation and professionalisation that builds on existing customary arrangements rather than replacing them. The objective is not to formalise irrigation management in a uniform or bureaucratic manner, but

to incrementally strengthen roles, capacities, and accountability mechanisms in ways that preserve local legitimacy while addressing emerging technical, financial, and climatic challenges.

Field consultations show that customary institutions—such as locally recognised water managers, elders' councils, and collective labour systems—continue to command authority and compliance in routine operation and maintenance. These institutions provide the social foundation upon which any form of institutional upgrading must rest. Comparative experience suggests that professionalisation is most effective when it focuses on functions rather than organisational form, for example by improving record-keeping, clarifying maintenance responsibilities, introducing basic asset management practices, or enabling access to skilled technical services, while retaining community control over decision-making (Annex C; Ostrom, Governing the Commons, 1990).

A recurring lesson from regional experience is that incremental institutional upgrading can enhance system performance without undermining social cohesion. In several contexts, community irrigation bodies evolved gradually to assume additional responsibilities—such as managing modest cash contributions, contracting skilled labour for complex repairs, or coordinating with public agencies—while remaining locally owned and accountable. These transitions were typically facilitated by enabling legal frameworks, technical backstopping, and learning-by-doing approaches rather than by the creation of entirely new institutions. Where formalisation was imposed rapidly or without sensitivity to local norms, outcomes were often negative, leading to disengagement or elite capture (World Bank, Sustaining Irrigation Services: Institutional and Financing Reforms, 2015).

Professionalisation of O&M also implies selective engagement with external expertise. Field evidence from Gilgit-Baltistan indicates that communities already rely on external technical support for complex works, particularly following flood or GLOF-related damage. Institutionalising such arrangements—through clearer contracting practices, transparent decision-making, and defined roles—can improve efficiency and accountability without transferring ownership or operational control. Comparative literature suggests that such community-led service arrangements, rather than conventional models, are better suited to low-revenue, high-risk environments (IFAD, Sustainable Smallholder Irrigation Development, 2016).

The role of the public sector in this process is primarily one of institutional stewardship. Rather than operating irrigation systems directly, public institutions can support professionalisation by recognising community entities, clarifying mandates, setting minimum technical and safety standards, facilitating access to training and technical advice, and linking communities to financing and disaster response mechanisms. This enabling role is particularly important in contexts like Gilgit-Baltistan, where more than 90 percent of irrigation infrastructure remains outside the formal sector and where community systems face increasing pressure from climate variability and demographic change (EU Policy Support Technical Assistance to Government of Gilgit-Baltistan, Inception Report, 2024).

Taken together, the evidence suggests that institutionalisation and professionalisation of O&M in Gilgit-Baltistan should be understood as a gradual strengthening of existing community systems, not as a transition toward externally operated or fully market-based models. Approaches that respect customary authority, introduce modest functional upgrades, and are supported by coherent public stewardship offer the most realistic pathway to improving irrigation sustainability while preserving community ownership and legitimacy.

6. Synthesis of Key Findings, Gaps, and Constraints

6.1 Key Findings

A central finding of the study is that irrigation in Gilgit-Baltistan is fundamentally a community-owned and community-managed system. More than 90 percent of traditional irrigation infrastructure has been developed, operated, and maintained through informal customary arrangements. These systems reflect long-standing adaptive responses to extreme environmental constraints, including low precipitation, pronounced seasonal variability, and reliance on snowmelt and glacial runoff. Indigenous engineering practices, collective labour mobilisation, and customary governance have enabled sustained agricultural production in an otherwise inhospitable environment.

The policy and institutional review indicate that, despite recent legislative and policy developments signalling increased recognition of community roles and environmental protection, formal institutional engagement with irrigation management remains limited. Public sector involvement is largely confined to capital investments, post-disaster rehabilitation, and project-based interventions, with minimal presence in routine operation and maintenance. This has resulted in a persistent disconnect between formal policy frameworks and the informal systems that continue to dominate irrigation management on the ground.

Infrastructure assessment findings show that system vulnerability is increasing across regions. Ageing channels, deferred maintenance, and heightened exposure to climate-related hazards—including floods, landslides, and glacial lake outburst floods—are placing growing stress on irrigation networks. While communities continue to mobilise labour and local resources for routine upkeep, the technical complexity and scale of damage increasingly exceed local capacity, leading to episodic reliance on external assistance.

Analysis of modern irrigation technologies suggests that limited adoption is driven less by lack of awareness or availability than by institutional and economic factors. In collective irrigation systems governed by customary allocation rules, individual investments in efficiency-enhancing technologies yield limited private returns. When combined with subsistence-oriented cropping patterns and the availability of subsidised staple food, incentives for farmers to invest in capital-intensive irrigation technologies remain weak. Where such technologies have been introduced, uptake has generally been partial, project-bound, and dependent on continued external support.

Operation and maintenance analysis demonstrates that customary O&M systems remain functional but are under increasing strain. Labour shortages linked to migration and livelihood diversification, rising cash requirements for technically complex repairs, and escalating climate risks are eroding the viability of purely voluntary maintenance arrangements. At the same time, evidence from both the field and comparative experience indicates that abrupt formalisation or externally imposed management models are poorly suited to the Gilgit-Baltistan context. More sustainable outcomes are associated with incremental strengthening of existing institutions rather than their replacement.

An additional system-level constraint identified through field evidence is the increasing seasonal mismatch between water availability and crop demand. Irrigation systems are exposed to early-season excess flows followed by mid- to late-season deficits, with limited capacity to store or regulate water across time. This results in sub-optimal water use, yield losses, and increased vulnerability during critical cropping periods. The absence of buffering mechanisms further amplifies the impact of climate variability on system performance.

Across all thematic areas, a consistent finding is that irrigation performance and long-term sustainability in Gilgit-Baltistan are shaped primarily by institutional arrangements, incentives, and collective capacity rather than by technical design alone. Interventions that overlook these structural dimensions are unlikely to achieve durable or scalable impact.

6.2 Key Gaps and Constraints

Despite the resilience and adaptive capacity of community-managed irrigation systems in Gilgit-Baltistan, the study identifies several persistent gaps and structural constraints that limit long-term sustainability. Foremost among these is the absence of systematic empirical data on irrigation system performance, maintenance costs, water losses, and financing flows. This data gap constrains the ability of both communities and public institutions to plan investments, prioritise interventions, or assess sustainability over time.

Institutional roles and mandates related to irrigation remain fragmented and weakly coordinated. While multiple public agencies engage with irrigation infrastructure at different stages—ranging from planning and construction to rehabilitation and disaster response—no coherent framework exists to support community systems across the full lifecycle of operation, maintenance, and adaptation to climate risk. This fragmentation is particularly evident in relation to technical backstopping, post-disaster recovery, and mechanisms for institutional learning.

Financing arrangements for irrigation operation and maintenance remain structurally constrained. Subsistence-oriented agriculture, limited cash incomes, and weak market integration reduce the feasibility of cost-recovery approaches based on user fees or tariffs. At the same time, the absence of predictable public financing mechanisms leaves communities exposed to recurrent shocks and reliant on ad hoc external assistance. This combination of limited household capacity and weak public stewardship perpetuates underinvestment in preventive maintenance and climate-resilient upgrading.

A further constraint arises from the persistent institutional gap between customary systems and formal governance frameworks. Although customary institutions continue to regulate water allocation and maintenance effectively at the local level, they operate largely outside formal recognition and support structures. This limits their access to technical training, financial services, and institutional development opportunities, and constrains prospects for gradual upgrading and professionalisation.

The study also highlights the limits of technology-led solutions in the absence of enabling conditions. Without aligned incentives, system-level coordination, and supportive institutional arrangements, modern irrigation technologies and digital tools are unlikely to transform irrigation performance at scale. In many cases, technology adoption risks increasing inequality or system fragmentation if introduced without attention to governance and collective action.

Taken together, these gaps and constraints indicate that the future sustainability of irrigation in Gilgit-Baltistan does not depend on wholesale system transformation. Rather, it hinges on carefully sequenced, institutionally grounded interventions that recognise the centrality of community systems while addressing their emerging technical, financial, and organisational limitations. This synthesis provides the analytical foundation for the strategic options and recommendations presented in the following section.

7. Implementation, Fiscal Context and Strategic Recommendations

7.1 Phased Implementation Pathway

Given the structural characteristics of irrigation systems in Gilgit-Baltistan, transformation must be approached through a sequenced and adaptive pathway rather than a single-stage reform model. The findings of this study indicate that sustainable improvements depend on aligning institutional strengthening, infrastructure support, and climate resilience measures over time.

In the short term (1–2 years), priority should be given to stabilizing existing systems through targeted rehabilitation of critical infrastructure, particularly intakes and vulnerable channel sections, combined with initial institutional strengthening measures. This includes formal recognition of community irrigation institutions, improved coordination mechanisms, and pilot initiatives for structured operation and maintenance support.

In the medium term (3–5 years), the focus should shift toward system strengthening and integration. This includes developing sustainable financing mechanisms for routine operation and maintenance, strengthening linkages between irrigation management and agricultural extension, and introducing context-appropriate technical improvements such as selective lining, protection works, and decentralized storage solutions. Institutional roles should be clarified and gradually operationalized, particularly at the interface between community systems and public sector agencies.

In the long term (5–10 years), the objective should be to transition toward climate-resilient irrigation systems that integrate hydrological variability, institutional sustainability, and technological adaptation. This includes scaling up storage and buffering systems, embedding irrigation within broader landscape and watershed management frameworks, and strengthening data systems for monitoring and adaptive management. At this stage, institutional arrangements should evolve toward hybrid models that combine customary governance with formal technical and financial support systems.

This phased approach reflects both the constraints and opportunities identified in the study and provides a realistic pathway for improving system performance without disrupting existing social and institutional foundations.

7.2 Public Financing Context

The implementation of recommended interventions must be considered within the fiscal context of Gilgit-Baltistan's Annual Development Programme (ADP), which reflects the primary source of public investment in irrigation infrastructure. Recent allocations to the irrigation sector have remained relatively stable, ranging between approximately PKR 190–270 million annually over the past four years. While this indicates continued government commitment, the scale of funding remains limited relative to the widespread infrastructure needs and increasing climate-related risks affecting irrigation systems.

Given these constraints, the findings of this study suggest that improvements in system performance cannot be achieved through large-scale expansion of infrastructure alone. Instead, there is a need to prioritise cost-effective, high-impact interventions focused on rehabilitation of critical assets, strengthening of operation and maintenance systems, and incremental resilience-building measures such as decentralised storage. Aligning investment priorities with system-level vulnerabilities and institutional realities will be essential to maximise the effectiveness of available public resources.

Table 14. ADP Allocation for the Irrigation Department

Year	Irrigation Allocation (PKR million)
2022–23	267
2023–24	190
2024–25	189
2025–26	192
Source: GB Irrigation and water management department	

7.3 Thematic Recommendations

This section presents strategic recommendations derived directly from the study's findings and synthesis. The recommendations focus on strengthening the sustainability, resilience, and inclusiveness of irrigation systems in Gilgit-Baltistan through incremental institutional upgrading, targeted public support, and locally grounded implementation mechanisms.

Theme 1: Upgrading Customary O&M Institutions into Community Water Management Utilities

The study identifies weakening operation and maintenance arrangements as the primary constraint to irrigation sustainability. The core strategic recommendation is therefore to upgrade existing customary water management institutions into community-owned Water Management Utility entities, without undermining local ownership, authority, or legitimacy.

Recommended actions include:

- Document customary water management systems in each valley, including locally recognised institutions, roles, allocation rules, maintenance practices, and conflict resolution mechanisms.
- Use this documentation as the basis for formally recognising customary institutions as the legitimate managers of community-owned irrigation systems.
- Where service charges or user contributions are locally agreed, upgraded institutions may adopt digital payment and record-keeping systems to strengthen financial transparency and accountability, while retaining community control over tariff setting and enforcement.
- Incrementally upgrade these institutions into community Water Management Utility entities, building on existing governance structures rather than replacing them.
- Mandate the upgraded utilities to manage integrated water services, including irrigation, drinking water supply, and basic sewerage, where these systems are interlinked.
- Clarify internal roles and responsibilities within utilities, including water allocation, routine maintenance, financial management, emergency response, and liaison with public agencies.
- Introduce basic asset management practices, such as inventories of channels, intakes, and high-risk sections, to support preventive maintenance and planning.

- Regulate these utilities through Local Governments and the Local Government and Rural Development Department, with regulation focused on standards, transparency, and public interest safeguards rather than operational control.

Theme 2: Public Funding and Technical Support as Institutional Enablers

Given the limited revenue base of subsistence-oriented agriculture, the study finds that sustainable institutional upgrading cannot rely on community resources alone. Strategic public funding and technical support are therefore essential enablers.

Recommended actions include:

- Establish dedicated public funding windows to support institutional upgrading, capacity development, and start-up costs for community Water Management Utilities.
- Prioritise public financing for rehabilitation of irrigation infrastructure damaged by floods, landslides, and glacial lake outburst events.
- Provide technical assistance focused on engineering diagnostics, climate-resilient design, and institutional development rather than routine operation.
- Ensure that public funding mechanisms are predictable, transparent, and linked to basic governance and performance criteria of the utilities.

Theme 3: Piloting, Learning, and Adaptive Scaling

To manage risk and avoid premature standardisation, the study recommends a pilot-driven approach that allows institutional models to be tested and refined.

Recommended actions include:

- Launch approximately ten pilot community Water Management Utilities, with one pilot in each district.
- Select pilot sites to reflect diversity in geography, irrigation system types, hazard exposure, and socio-economic conditions.
- Use pilots as structured learning platforms to test governance arrangements, cost-sharing mechanisms, service integration, and regulatory oversight.
- Document lessons systematically and use them to inform gradual scaling and policy refinement.

Theme 4: Embedding Social Safeguards and Inclusive Governance

Demographic change, migration, and evolving social roles require irrigation institutions to become more inclusive to remain sustainable.

Recommended actions include:

- Require upgraded O&M institutions and community utilities to include women and youth in defined functional roles.
- Prioritise women's and youth participation in areas such as financial management, record-keeping, monitoring, and coordination with external agencies.
- Provide targeted training and capacity support to ensure participation is substantive rather than symbolic.

- Incorporate inclusion benchmarks into regulatory oversight and performance monitoring by local governments.

Theme 5: Aligning Irrigation Reform with Broader Development Objectives

The study finds that irrigation sustainability is closely linked to food security, public health, and environmental outcomes.

Recommended actions include:

- Align irrigation upgrading with improvements in drinking water supply and sanitation to maximise health and environmental co-benefits.
- Coordinate irrigation support with agricultural extension, crop diversification, and market access initiatives.
- Frame irrigation reform as a contributor to broader development objectives, including poverty reduction, inclusive institutions, and sustainable resource management.

8. Legal and Policy Reform Agenda

This section sets out a focused legal and policy reform agenda required to support the strategic recommendations of this study, particularly the upgrading of customary operation and maintenance institutions into community-owned Water Management Utility entities. The emphasis is on enabling, clarifying, and safeguarding existing community systems rather than creating parallel or replacement structures. The proposed reforms are incremental and designed to align formal frameworks with the realities of irrigation management in Gilgit-Baltistan.

8.1 Formal Recognition of Customary Water Management Institutions

A central legal gap identified by the study is the absence of explicit recognition of customary water management institutions within formal policy and regulatory frameworks. While these institutions manage the majority of irrigation systems, their informal status limits access to public support, technical assistance, and financing.

Required actions include:

- Introducing legal or regulatory provisions that recognise customary irrigation institutions as legitimate local service entities responsible for operation and maintenance of community-owned systems.
- Allowing documented customary institutions to interface formally with public agencies without requiring full corporatisation or conversion into standardised entities.
- Ensuring that recognition frameworks preserve community ownership, authority, and internal decision-making arrangements.

8.2 Enabling Frameworks for Community Water Management Utility Companies

To support the proposed transition toward upgraded community Water Management Utility entities, the legal framework must allow flexible, locally adapted institutional forms.

Required actions include:

- Providing legal space for community-owned utility entities that combine irrigation, drinking water supply, and basic sewerage services where appropriate.
- Defining minimum governance and transparency requirements while avoiding rigid organisational templates.
- Allowing utilities to manage modest financial resources, enter service contracts, and receive public funding without undermining community control.

8.3 Clarification of Public Sector Stewardship Roles

The study finds that public sector engagement in irrigation is fragmented and largely project-driven. Legal and policy clarity is needed to reposition public institutions as stewards rather than operators.

Required actions include:

- Clarifying the mandates of Local Governments and the Local Government and Rural Development (LG&RD) Department in regulating, supporting, and overseeing community water management utilities.

- Defining stewardship functions, including setting minimum technical standards, supporting institutional development, facilitating training, and coordinating post-disaster response.
- Avoiding legal provisions that shift routine operation and maintenance responsibility to the public sector, which would be fiscally and operationally unsustainable.

8.4 Financing and Public Support Instruments

Existing policy frameworks provide limited guidance on financing operation and maintenance of community-managed irrigation systems.

Required actions include:

- Establishing policy provisions that allow targeted public funding for institutional upgrading, technical backstopping, and rehabilitation following natural disasters.
- Linking public support to basic governance and performance criteria rather than cost recovery targets that are unrealistic in subsistence-oriented contexts.
- Ensuring that financing instruments are predictable and rules-based to reduce reliance on ad hoc project interventions.

8.5 Embedding Social Safeguards and Inclusive Governance

The study highlights the need to address demographic change and inclusion through institutional reform.

Required actions include:

- Incorporating provisions that encourage participation of women and youth in upgraded water management institutions and utility entities.
- Requiring transparency and inclusiveness in governance arrangements as part of recognition and regulatory oversight.
- Aligning local reforms with broader national and international commitments related to inclusive institutions, equity, and sustainable development.

8.6 Learning-Oriented Legal Adaptation

Given the diversity of irrigation systems and institutional arrangements across Gilgit-Baltistan, the study cautions against premature standardisation.

Required actions include:

- Allowing pilot initiatives to operate under adaptive legal arrangements where necessary.
- Using lessons from district-level pilots to refine legal and policy provisions over time.
- Treating legal reform as an iterative process informed by evidence and implementation experience rather than a one-time codification exercise.

8.7 Monitoring and Evaluation Framework

To support effective implementation of the proposed legal and policy reform agenda, a dedicated Monitoring and Evaluation (M&E) Framework has been prepared as part of this study. The framework is designed to track progress in legal recognition, institutional strengthening, public

sector stewardship, financing arrangements, and inclusive governance, in line with the reform areas set out in this section.

The M&E Framework emphasises monitoring of institutional and policy change processes rather than short-term infrastructure outputs. Given the incremental and adaptive nature of the proposed reforms, the framework focuses on tracking legal and regulatory milestones, institutional practices, and the functioning of stewardship and support mechanisms. This approach is intended to support learning, accountability, and evidence-based adjustment of reform measures over time.

Consistent with the objectives of the Policy Support Technical Assistance, the framework adopts a learning-oriented and adaptive monitoring approach. It allows for periodic review of pilot initiatives, assessment of reform effectiveness, and refinement of legal and policy provisions based on implementation experience. Monitoring responsibilities are aligned with existing government mandates to minimise reporting burdens and encourage integration into routine governance processes.

Annexures

Annex A: Literature Review

Introduction

This annex presents a targeted review of literature relevant to irrigation systems in Gilgit-Baltistan, prepared in support of the EU Policy Support Technical Assistance to the Government of Gilgit-Baltistan. The purpose of the literature review is to provide an analytical foundation for the main report by situating field-based findings within established policy frameworks, documented institutional experience, and relevant academic research. It supports policy interpretation and option development rather than serving as a standalone academic review.

The literature reviewed serves three interrelated functions. First, it provides contextual grounding on the environmental, institutional, and governance settings within which irrigation systems in Gilgit-Baltistan operate. Second, it offers explanatory insights from mountain and Karakoram-focused studies that assist in interpreting observed patterns related to infrastructure design, water allocation, operation and maintenance, and institutional performance (Kreutzmann, 1996; 2011). Third, it highlights structural constraints and enabling conditions that shape irrigation sustainability and reform trajectories.

The scope of the review is deliberately bounded. Priority is given to literature directly relevant to high-mountain irrigation systems, with emphasis on Gilgit-Baltistan and closely comparable Karakoram and Pamir contexts. Sources include sub-national legislation, policy and planning documents, peer-reviewed academic studies, and institutional publications by recognised research and development organisations active in Northern Pakistan.

Primary data generated through field consultations under this study are analysed in the main report and are not treated as literature. This Annex Aocuses exclusively on secondary sources and documented experience, maintaining a clear distinction between empirical findings and prior knowledge.

Methodology for the Literature Review

The literature review followed a selective and policy-oriented approach aligned with EU Policy Support Technical Assistance standards. Emphasis was placed on relevance, applicability, and analytical value rather than comprehensive academic coverage.

Sources were selected based on their relevance to mountain irrigation systems, their applicability to Gilgit-Baltistan, and their contribution to understanding institutional, legal, governance, environmental, and sustainability dimensions of irrigation. Literature was grouped into five categories: (i) legal and policy frameworks; (ii) academic literature on mountain irrigation and water governance; (iii) institutional and programme documentation; (iv) customary law and community-based governance studies; and (v) climate, risk, and environmental protection literature.

Limitations are acknowledged. Dedicated irrigation policy documents specific to Gilgit-Baltistan remain limited, and several academic studies draw on fieldwork conducted over the past two decades. Empirical evidence on irrigation financing, performance monitoring, and institutional effectiveness remains relatively sparse, informing a cautious and evidence-bounded interpretation.

Irrigation Systems in Gilgit-Baltistan: Physical, Hydrological, and Typological Context

Available literature indicates that irrigation systems in Gilgit-Baltistan are shaped by steep topography, climatic variability, and dependence on glacier and snowmelt-fed hydrology (Kreutzmann, 1996; 2011). In contrast to lowland canal systems, irrigation in the Karakoram operates as a tightly coupled socio-ecological system, where physical infrastructure, settlement patterns, and collective institutions are interdependent.

Seasonal variability in flows, sediment loads, and channel instability place structural constraints on design and operation. National water policy frameworks acknowledge variability and scarcity but are largely oriented toward basin-scale river systems, offering limited operational guidance for small-scale mountain irrigation networks (Government of Pakistan, 2018; PCRWR, 2023).

Typologically, the literature distinguishes between community-managed systems, government-supported systems integrated into traditional layouts, and fully public schemes. Evidence suggests that system performance is more closely associated with governance arrangements, maintenance capacity, and collective action than with technical design alone (Kreutzmann, 2011).

Policy, Legal, and Institutional Frameworks for Irrigation

Irrigation systems in Gilgit-Baltistan operate within a policy and institutional environment that is evolving but remains partially aligned with mountain realities. At the national level, instruments such as the National Water Policy and the National Water Conservation Strategy articulate principles of efficiency, equity, and sustainability, but are primarily framed around large river basins and command-area irrigation (Government of Pakistan, 2018; PCRWR, 2023).

At the sub-national level, recent legislative developments represent a significant shift toward formal recognition of community-based water governance. The Gilgit-Baltistan Water User's Associations Act, 2024 establishes a statutory framework for the formation, registration, and operation of Water User Associations (WUAs), formally recognising their role in irrigation maintenance, cost-sharing, dispute resolution, and collective management. The Act provides legal personality to WUAs, clarifies relations with government field officers, and introduces enforceable mechanisms for cost recovery and compliance.

Planning instruments, particularly Annual Development Programmes, continue to prioritise infrastructure development, with comparatively limited emphasis on long-term operation and maintenance financing. Literature suggests that the effectiveness of the new WUA framework will depend on how it is operationalised, resourced, and aligned with customary practices (Bilal et al., 2003; Kreutzmann, 2011).

Customary Law and Community-Based Irrigation Governance

A consistent finding across the literature is the central role of customary institutions in irrigation governance in Gilgit-Baltistan. These institutions regulate water allocation, mobilise labour for maintenance, and resolve disputes through socially legitimate mechanisms (Bilal et al., 2003; Kreutzmann, 1996).

Customary irrigation governance in Gilgit-Baltistan has been formally documented through early institutional efforts to codify indigenous natural resource management systems, including water rights, labour obligations, and sanctions (IUCN Pakistan, 2000).

Customary systems typically define water rights through rotational shares linked to landholding and labour contribution. Enforcement relies on social accountability rather than formal sanctions. The literature cautions that externally imposed institutional models that fail to recognise these arrangements risk undermining system functionality.

The Water User's Associations Act, 2024 provides a formal interface between customary governance and the state. By allowing WUAs to register, hold bank accounts, and recover costs, the Act offers a pathway for institutional strengthening, provided that statutory requirements are applied flexibly and in harmony with existing social norms.

Infrastructure Condition, Risk, and Environmental Regulation

Mountain irrigation infrastructure in Gilgit-Baltistan is exposed to multiple interacting risks, including floods, landslides, channel erosion, and glacial lake outburst floods (Kreutzmann, 2011). Climate-related studies indicate increasing variability in flows and sediment dynamics, amplifying maintenance demands (ICIMOD, various years).

The Gilgit-Baltistan River Protection Act, 2024 introduces an important environmental and regulatory dimension by establishing floodplain zoning, construction controls, and pollution prohibitions along rivers, streams, and nullahs. The Act requires no-objection certificates for infrastructure works near water bodies and mandates sanitation and waste management controls in riparian zones.

From an irrigation perspective, the Act has implications for the siting, rehabilitation, and protection of irrigation intakes, channels, and crossings. Literature suggests that effective implementation will require coordination between environmental authorities, irrigation departments, and community institutions to avoid regulatory fragmentation and unintended constraints on traditional water systems.

Operation, Maintenance, and Institutional Sustainability

Operation and maintenance are widely recognised as critical determinants of irrigation sustainability in mountain environments. Traditional systems rely on seasonal maintenance cycles, collective labour mobilisation, and locally enforced compliance (Bilal et al., 2003).

Demographic change, labour migration, and increasing climate-related damage have strained these arrangements (Kreutzmann, 2011). Public sector involvement in routine maintenance remains limited, with a continued emphasis on capital investment over recurrent expenditure.

The Water User's Associations Act, 2024 introduces mechanisms for cost recovery, collective responsibility, and state-backed enforcement, potentially strengthening O&M sustainability if supported through technical assistance and predictable funding.

Irrigation Modernization and Institutional Change

Literature on mountain irrigation cautions against equating modernisation with standardised engineering solutions. Evidence from the Karakoram suggests that incremental, context-specific improvements—such as selective lining, improved intakes, and stabilisation of vulnerable reaches—are more effective when embedded within existing layouts and governance arrangements (Kreutzmann, 2011).

Institutional change is inseparable from technical change. The combination of statutory WUAs and strengthened environmental regulation creates opportunities for more integrated water governance, provided that implementation remains adaptive and locally grounded.

Social Dimensions and Equity Considerations

Social dimensions of irrigation governance receive limited attention in formal policy documents but are highlighted in institutional and academic literature. Gendered roles in agriculture and water management are well documented, with women contributing significantly to production and on-farm water use but often under-represented in decision-making bodies (IUCN Pakistan, various years).

Youth outmigration and livelihood diversification reduce labour availability and weaken intergenerational transmission of technical and institutional knowledge, with implications for long-term system sustainability (Kreutzmann, 2011).

Comparative Experience from Other Mountain and Communal Irrigation Systems

Comparative literature from other mountain and communal irrigation contexts provides useful reference points for interpreting the findings from Gilgit-Baltistan, while also highlighting the limits of direct transferability. Evidence from Farmer-Managed Irrigation Systems (FMIS) in Nepal, Bhutan, and parts of India consistently demonstrates that irrigation performance in small-scale mountain systems is more strongly influenced by collective governance arrangements than by the adoption of individual efficiency technologies. Studies show that where water allocation is governed through rotational or proportional sharing, individual investments in efficiency-enhancing technologies do not result in private gains unless system-level rules are modified to internalise water savings (Ostrom, 1992; Pradhan, 2010).

Research from Central Asia following the post-Soviet transition further illustrates the institutional dimensions of irrigation modernisation. The introduction of Water User Associations in Kyrgyzstan, Tajikistan, and Uzbekistan aimed to improve efficiency and cost recovery, but outcomes have been mixed. Evidence indicates that where WUAs were introduced as formal entities without strong social legitimacy or incentive alignment, technology adoption and maintenance outcomes remained weak despite infrastructure investment (Sehring, 2009; Abdullaev et al., 2010). These findings echo challenges observed in Gilgit-Baltistan, where formal institutional frameworks must interface carefully with customary governance systems to be effective.

Experience from the Andean region in Latin America similarly underscores the centrality of collective action and locally embedded norms. Studies of communal irrigation systems in Peru and Bolivia show that efficiency-oriented technologies and flow measurement tools have limited impact unless they are embedded within recognised community governance structures and accompanied by negotiated changes in allocation rules (Boelens, 2015). In these contexts, attempts to introduce modern technologies without institutional adaptation have often led to conflict, underutilisation, or abandonment.

Across these regions, the literature converges on a common finding: modern irrigation technologies and digital tools function effectively only as complements to strong institutions, shared incentives, and collective decision-making processes. Where agriculture remains subsistence-oriented and markets are weak, technology adoption tends to remain limited regardless of technical feasibility. These comparative insights reinforce the GB-specific findings of this study and caution against uncritical application of models derived from fundamentally different agroeconomic and governance environments.

Institutionalisation and Professionalisation of Operation and Maintenance

Comparative literature on small-scale and community-managed irrigation systems highlights the limits of purely voluntary, labour-based operation and maintenance arrangements under conditions of increasing system complexity, climate stress, and socio-economic change. While customary institutions have historically sustained irrigation infrastructure through collective labour and social enforcement, evidence from multiple regions indicates that such arrangements become increasingly strained as maintenance demands rise, infrastructure ages, and labour availability declines due to outmigration and livelihood diversification.

Experience from Central Asia and parts of Latin America shows that one pathway for improving long-term sustainability has been the gradual institutionalisation of O&M functions through locally owned or community-based service entities. In these cases, Water User Associations or similar bodies evolved beyond informal coordination roles to assume responsibility for routine maintenance, fee collection, contracting of technical services, and basic asset management. Where such entities retained strong local ownership and accountability, professionalisation contributed to improved system reliability without undermining social legitimacy.

The literature emphasises that successful transitions toward semi-formal or utility-style arrangements depend on several enabling conditions. These include clear definition of roles between communities and government agencies, predictable financing mechanisms for recurrent costs, access to technical backstopping, and legal frameworks that recognise community entities as service providers rather than merely beneficiaries. Where formalisation was imposed without sensitivity to existing norms and power structures, outcomes were often negative, resulting in disengagement, elite capture, or institutional hollowing.

These insights are directly relevant to Gilgit-Baltistan in light of recent legislative developments, including the Water User's Associations Act, 2024. Comparative evidence suggests that long-term sustainability of irrigation systems is more likely to be strengthened through incremental upgrading of customary management, maintenance, and governance practices—toward professionally supported, locally owned service arrangements—than through abrupt replacement of indigenous institutions with externally designed utility models. This perspective provides an important analytical bridge between the literature review and the operational realities examined in the main report.

Synthesis of Key Insights and Gaps

Taken together, the literature indicates that irrigation systems in Gilgit-Baltistan represent adaptive responses to highly constrained mountain environments, shaped by seasonal water variability, geomorphological risk, and limited state presence. These systems are sustained primarily through collective governance arrangements, customary rules, and incremental technical adjustments rather than through standardised engineering or market-driven intensification. This pattern is consistent with evidence from other communal and mountain irrigation contexts, reinforcing the central role of institutions and incentives in determining system performance.

Comparative experience from South Asia, Central Asia, and Latin America underscores that irrigation efficiency, technology adoption, and sustainability outcomes in communal systems depend less on the availability of modern technologies and more on governance arrangements, incentive structures, and coordination mechanisms. Across regions, efficiency-oriented technologies and digital tools have demonstrated limited impact unless embedded within

recognised institutions, coordinated decision-making processes, and financing arrangements that internalise system-level benefits. These insights validate the relevance of the Gilgit-Baltistan experience while cautioning against uncritical transfer of models developed in fundamentally different agro-economic and governance settings.

Recent legislative developments in Gilgit-Baltistan, including the Water User's Associations Act, 2024 and the River Protection Act, 2024, represent important steps toward formal recognition of community roles and strengthening environmental safeguards. However, the literature also indicates that legal recognition alone is insufficient to improve performance. Without clarity of mandates, sustainable financing for operation and maintenance, and effective integration between customary institutions and formal administrative systems, the practical impact of new legal frameworks may remain limited.

Significant gaps remain in the empirical evidence base. Existing literature provides limited insight into actual irrigation system performance over time, including reliability, equity of water distribution, maintenance effectiveness, and resilience under increasing climate stress. Evidence on financing arrangements, institutional effectiveness at different scales, and the real-world impacts of modern irrigation technologies and digital tools in communal settings remains fragmented. These gaps reinforce the importance of the field-based analysis presented in the main report, which grounds policy and institutional recommendations in observed practices, constraints, and incentives rather than normative assumptions.

Annex B: Field Instruments and Research Tools

Purpose of Annex

This annex documents the field instruments and research tools used to generate the primary evidence base for the Irrigation Sector Study conducted under the European Union–supported Policy Support Technical Assistance to the Government of Gilgit-Baltistan. The instruments were designed to support structured, comparable, and traceable qualitative data collection across regions and stakeholder groups, and to ensure consistency between field evidence, analytical findings, and policy recommendations.

Field Work Strategy

The field work strategy was designed to capture institutional, technical, operational, and governance dimensions of irrigation systems across diverse ecological and socio-institutional contexts in Gilgit-Baltistan. The strategy defined:

- Regional coverage, including Gilgit, Baltistan, and Diamer/Astore
- Stakeholder categories to be consulted
- Consultation modalities and sequencing
- Procedures for documentation, validation, and synthesis of field evidence

The strategy emphasised triangulation across stakeholder perspectives, regions, and secondary sources, and prioritised qualitative depth over statistical representation. Fieldwork was conducted with attention to seasonal conditions, accessibility, and local institutional arrangements.

Stakeholder Categories

Primary evidence was collected through structured consultations with the following stakeholder groups:

- Farmers and water users directly dependent on irrigation systems
- Customary water management actors, including elders, mirabs, and water committees
- Community representatives involved in collective decision-making
- Local Government representatives and relevant line departments
- Institutions engaged in irrigation support, rehabilitation, or disaster response

Stakeholder selection aimed to capture both operational experience and institutional perspectives across the irrigation system lifecycle.

Consultation Instruments

Structured consultation instruments were developed to guide discussions and ensure consistency across regions and stakeholder groups. The instruments comprised thematic question sets organised under the following domains:

Irrigation System Characteristics

- Source of irrigation water and system configuration
- Scale, command area, and infrastructure condition
- Key vulnerabilities and exposure to natural hazards

Operation and Maintenance Practices

- Responsibility for routine and emergency maintenance
- Labour mobilisation mechanisms
- Technical capacity for repairs and system upkeep

Institutional Arrangements and Governance

- Decision-making processes for water allocation and maintenance
- Role of customary institutions and leadership structures
- Interaction with formal public institutions

Financing and Cost-Sharing

- Use of collective labour versus monetary contributions
- Circumstances under which cash contributions are mobilised
- Experience with external funding or project support

Equity, Inclusion, and Conflict Resolution

- Access to water across households and groups
- Participation of women and youth in irrigation governance

- Mechanisms for resolving disputes related to water use

Climate Risks and System Resilience

- Observed climate-related impacts on irrigation systems
- Coping and adaptation strategies at community level
- Support received during extreme events

Questions were open-ended and designed to elicit descriptive evidence, institutional insights, and experiential learning rather than binary responses.

Documentation, Coding, and Traceability

Responses from consultations were recorded, expanded into narrative form, and systematically coded using a structured reference system. Each response was linked to:

- A specific question or thematic domain
- Stakeholder category
- Region
- Sequential reference number

This coding framework underpins Annex C and enables traceability from synthesised findings back to original field evidence and transcripts. The approach supports transparency, internal consistency, and analytical validation while preserving the qualitative nature of the data.

Quality Assurance Measures

Quality assurance measures were applied throughout the data collection and analysis process, including:

- Cross-checking responses across stakeholder groups
- Comparative review of findings across regions
- Consistency checks against consultation instruments
- Triangulation with secondary sources and policy documents

These measures were intended to reduce anecdotal bias and strengthen the credibility of qualitative findings.

Limitations of the Field Instruments

The field instruments were designed to support qualitative, policy-oriented analysis rather than statistically representative sampling or engineering-level assessment. They do not provide quantitative measurements of water flows, system efficiency, or financial performance at scheme level. These limitations are consistent with the scope of the Technical Assistance and are addressed through triangulation and comparative analysis.

Linkage to Analytical Annexes

The application of the field instruments directly informs the synthesis presented in the main report and is operationalised through Annex C (Summary of Primary Evidence from the Field),

where coded responses are presented in structured form. Together, Annex A and Annex C provide a transparent chain from data collection to analysis and policy recommendation.

Annex C: Summary of Evidence Across all Domains of Enquiry

***Methodological Note**:*

This annex presents primary field evidence collected through structured consultations. Each entry links a specific field question to an expanded verbatim response, referenced through a unique code that enables backward tracing to full transcripts and field notes. The annex is designed to support transparent citation, triangulation, and synthesis in the main report. Following is the summary of all three regions. Annex C.1-C.3 provide region-wise summaries of primary evidence.

Domain 1 – Institutional and Governance Arrangements

Gilgit Region (C.1)

Irrigation governance in Gilgit is anchored in deeply embedded customary arrangements led by elders and respected community figures. Water rights are largely egalitarian at the household level, with little differentiation based on wealth or social status. Decision-making is consultative, and compliance is ensured through social norms rather than formal enforcement. Absentee landowners remain institutionally connected to the system and are expected to contribute either labour or cash. Conflicts are infrequent and are typically resolved through mediation by elders, reinforcing trust in the system.

Baltistan Region (C.2)

Baltistan exhibits a similar reliance on customary governance, but with water allocation more explicitly linked to land ownership. Elders and water associations oversee allocation, scheduling, and dispute resolution through consensus-based mechanisms. While irrigation shares vary by land size, drinking water is treated as a collective right. Non-resident landowners are financially integrated into the system. Overall, governance arrangements are perceived as legitimate, transparent, and socially cohesive, with disputes emerging mainly during periods of acute scarcity.

Diamer / Astore Region (C.3)

In Diamer and Astore-type upper catchment contexts, governance is also customary but more rigidly structured around landholding. Dedicated water committees supervise distribution, reflecting higher stress on the system. Irrigation turns are proportional to land size, while drinking water remains equally shared. Absentee landowners are required to contribute cash at locally defined wage rates. Authority rests on trust and continuity rather than written rules, but governance pressure is visibly higher due to recurring shocks and infrastructure fragility.

Domain 2 – Infrastructure Quality, Operation and Maintenance

Gilgit Region (C.1)

Irrigation infrastructure consists primarily of long, earthen channels constructed through manual labour. These channels are exposed to erosion, leakage, and collapse. Annual collective maintenance is compulsory, with labour contributions enforced socially. Minor repairs are handled locally using traditional techniques, while major failures require external assistance that is often delayed. Government rehabilitation support exists but is uneven and frequently misaligned with mountain conditions, resulting in recurring damage.

Baltistan Region (C.2)

Baltistan's irrigation channels are similarly ancestor-built and earthen, often located on steep and unstable slopes. Regular communal maintenance remains the backbone of system upkeep,

but repeated flood damage increasingly overwhelms local capacity. Government interventions, including lining and rehabilitation, are selective and constrained by budgets. Communities recognise that current maintenance practices are no longer sufficient to cope with escalating physical stress on infrastructure.

Diamer / Astore Region (C.3)

Infrastructure conditions are most fragile in Diamer and Astore-type areas. Channels frequently traverse sandy and unstable terrain, leading to high seepage losses. Temporary fixes such as grass filling are common but ineffective over longer distances. Maintenance obligations are strictly enforced at household level, yet structural weaknesses persist. Government departments favour earthen solutions over cement due to altitude constraints, but limited funding restricts coverage and sustainability.

Domain 3 – Water Allocation, Equity and Access

Gilgit Region (C.1)

Water allocation follows a rotational system that prioritises equity across households. All households, including land-poor or landless ones, are guaranteed drinking water access. During scarcity, irrigation turns are shortened and monitored more closely, with elders intervening to prioritise essential crops and domestic needs. The system is widely viewed as fair and socially binding.

Baltistan Region (C.2)

In Baltistan, irrigation water is allocated proportionally to landholding, while drinking water is shared equally. During normal flow conditions, water often runs continuously, reducing the need for strict scheduling. Scarcity triggers tighter rotation and oversight by water committees. Despite unequal irrigation shares, the system is broadly accepted due to its transparency and historical legitimacy.

Diamer / Astore Region (C.3)

Allocation in Diamer is strictly land-based for irrigation, reflecting higher competition and limited channel capacity. Drinking water is treated separately and shared equally. Perceived inequities arise not from governance rules but from infrastructural constraints that prevent water from reaching all fields during peak demand periods. The system is seen as fair in principle but constrained in practice.

Domain 4 – Climate Stress, Risk and Environmental Change

Gilgit Region (C.1)

Communities report increasing frequency of floods and landslides, disrupting irrigation for weeks at a time. Glacier and snowmelt flows are highly seasonal and increasingly unpredictable, affecting planting calendars and yields. Preventive measures are largely ad hoc, relying on temporary embankments and emergency repairs rather than systematic protection.

Baltistan Region (C.2)

Baltistan faces similar climate stresses, with summer floods and landslides repeatedly damaging intakes and channels. Variability in snowmelt timing is disrupting traditional agricultural rhythms. Communities acknowledge that existing coping mechanisms are insufficient and lack both resources and technical guidance to implement climate-resilient infrastructure.

Diamer / Astore Region (C.3)

Climate impacts are most severe in Diamer. Annual floods, particularly in June and July, can shut down irrigation for 20–22 days, causing complete crop failure in some years. Communities have resorted to water tankers for drinking water during crises. There are no formal preventive or adaptive systems, and responses remain reactive.

Domain 5 – External Support, Innovation and Future Needs

Gilgit Region (C.1)

External support from NGOs and projects has helped rehabilitate some channels but has not addressed long-term operation and maintenance. Informal digital tools such as mobile phones and WhatsApp are increasingly used for coordination. Communities consistently call for durable intake protection, technically appropriate designs, and reliable emergency financing.

Baltistan Region (C.2)

Support from government and NGOs has been sporadic and short-term. WhatsApp and mobile payments are used for coordination and cash collection, particularly involving absentee landowners. Communities seek sustained government engagement, technical guidance tailored to mountain contexts, and institutionalised mechanisms for post-disaster response.

Diamer / Astore Region (C.3)

External assistance has been minimal, limited largely to crisis response by programmes such as ETI and GBRSP. Technological adoption is extremely limited beyond basic communication tools. Communities strongly emphasise the need for proper channel construction, advance emergency funds, and closer, more predictable coordination between government, communities, and potential private partners.

Annex C.1: Summary of Primary Evidence: Gilgit Region

C.1.1 Domain 1 – Institutional and Governance Arrangements

Question Asked	Verbatim Response	Reference
How are water rights allocated and enforced in your community?	Water is distributed equally among households according to a long-established customary system. Elders appoint specific individuals to manage turns and supervise distribution. There is no distinction between rich and poor households in access to water for irrigation or domestic use. Compliance is ensured through social norms rather than written rules.	(Hopar, Nagar) – GIL–FARM–D1–01
Who manages irrigation and how are decisions made?	A group of respected elders oversees irrigation management. They collectively decide on water turns, resolve disputes, and mobilise labour for maintenance. Decisions are taken through consultation and are rarely challenged, as the system is widely trusted.	(Danyor, Gilgit) – GIL–CBI–D1–02
How are absentee or non-resident landowners treated in the system?	Absentee landowners are still considered part of the irrigation system. They are expected to contribute either by arranging labour or by paying cash equivalent to the local daily wage. Failure to contribute is socially noted and may affect future cooperation.	(Juglot, Gilgit) – GIL–FARM–D1–03

C.1.2 Domain 2 – Infrastructure Quality, Operation and Maintenance

Question Asked	Verbatim Response	Reference
Who decides on canal cleaning, maintenance, and labour arrangements?	Canal cleaning and maintenance are organised annually before the irrigation season. Each household is required to send at least one able-bodied member. If someone cannot participate physically, they must contribute cash. The schedule is announced by elders and followed collectively.	(Hopar, Nagar) – GIL–FARM–D2–01
What are the most common technical problems and how are they addressed?	Channels are long, exposed, and often unlined, making them vulnerable to leakage, erosion, and collapse. Minor damages are repaired locally using stones, soil, and wooden supports. Larger failures require external assistance, which is not always available on time.	(Gilgit City Periphery) – GIL–FARM–D2–02
What role do government departments play in rehabilitation?	Government support is limited and mostly reactive. Some lining and rehabilitation works have been undertaken, but coverage is uneven. Communities report that technical designs often do not fully match mountain conditions, leading to repeated damage.	(Gilgit) – GIL–GOV–D2–03

C.1.3 Domain 3 – Water Allocation, Equity and Access

Question Asked	Verbatim Response	Reference

How is irrigation water distributed among households?	Water distribution follows a rotational system where each household receives a fixed turn. The system is perceived as equitable because all households receive water regardless of economic status. Any attempt to take excess water is socially discouraged.	(Hopar, Nagar) – GIL–FARM–D3–01
How is water managed during periods of scarcity?	During low-flow periods, irrigation turns are shortened and strictly monitored. Elders may temporarily prioritise essential crops and domestic needs. Disputes increase slightly during scarcity but are generally resolved through mediation.	(Danyor, Gilgit) – GIL–FARM–D3–02
How is drinking water treated within the irrigation system?	Drinking water is treated as a collective necessity. Even households with no agricultural land are entitled to sufficient water for domestic use. This principle is strongly upheld by the community.	(Juglot, Gilgit) – GIL–CBI–D3–03

C.1.4 Domain 4 – Climate Stress, Risk and Environmental Change

Question Asked	Verbatim Response	Reference
Has your irrigation system been affected by floods, landslides, or climate events?	Floods and landslides frequently damage intakes and channel sections, particularly during peak summer months. In some cases, irrigation is disrupted for several weeks, affecting crop yields and household food security.	(Gilgit District) – GIL–FARM–D4–01
What are the main sources of water and how reliable are they?	The main water sources are glacier and snowmelt. Water availability is highly seasonal, with shortages in early spring and late summer. Communities report increasing unpredictability in flows compared to the past.	(Nagar Valley) – GIL–FARM–D4–02
What measures have communities taken to manage climate risks?	Preventive measures are limited to temporary embankments, stone pitching, and community-led repairs. There is no systematic protection of intakes or channels against extreme events.	(Danyor, Gilgit) – GIL–CBI–D4–03

C.1.5 Domain 5 – External Support, Innovation and Future Needs

Question Asked	Verbatim Response	Reference
What external support have you received for irrigation?	Some support has been provided by NGOs and development programmes, mainly for channel rehabilitation and lining. However, coverage is partial and does not extend to long-term operation and maintenance.	(Gilgit) – GIL–NGO–D5–01
Have you adopted any new technologies or coordination tools?	Communities increasingly use mobile phones and informal WhatsApp groups to coordinate maintenance activities and inform members about water turns. No advanced monitoring or automation tools are in use.	(Hopar, Nagar) – GIL–FARM–D5–02
What support would most strengthen your irrigation system?	Respondents emphasised the need for durable intake protection, technically appropriate designs for mountain terrain, and a reliable mechanism for emergency funding after floods or landslides.	(Nagar Valley) – GIL–FARM–D5–03

Annex C.2: Summary of Primary Evidence: Baltistan Region

C.2.1 Domain 1 – Institutional and Governance Arrangements

Question Asked	Verbatim Response	Reference
How are water rights allocated and enforced in your community?	Water distribution in the village follows a long-established customary system rooted in land ownership and collective agreement. Elders and a water committee oversee allocation, ensuring that each household receives water proportional to its landholding. Drinking water is treated as a shared right, with equal access regardless of land size. Decisions are taken through consensus, and social pressure rather than formal sanctions ensures compliance.	(Skardu, Baltistan) – BAL–FARM–D1–01
Who manages irrigation and how are disputes resolved?	There is no written law governing irrigation, but the authority of elders and the water association is widely respected. Disputes are resolved informally through dialogue, often involving senior community members. The system is perceived as fair, and conflicts are rare unless water scarcity becomes severe.	(Khaplu, Baltistan) – BAL–CBI–D1–02
How do absentee or non-resident landowners contribute to irrigation management?	Non-resident landowners remain part of the irrigation governance system. While they do not participate physically in maintenance, they are expected to contribute financially based on prevailing local wage rates. Failure to contribute is socially discouraged and may result in community pressure or delayed access to water.	(Shigar, Baltistan) – BAL–FARM–D1–03

C.2.2 Domain 2 – Infrastructure Quality, Operation and Maintenance

Question Asked	Verbatim Response	Reference
Who decides on canal cleaning, maintenance, and labour arrangements?	The irrigation channels were constructed by ancestors using manual labour and simple tools. Most channels are earthen, running along steep mountain slopes and exposed to erosion. Annual maintenance is compulsory; each household must either send one able person for cleaning and repair or provide cash if unable to do so.	(Kharmang, Baltistan) – BAL–FARM–D2–01
What are the most common technical problems and how are they addressed?	Floods and landslides repeatedly damage intakes and channel sections, particularly during June and July. When damage exceeds local capacity, temporary repairs are made to restore minimal flow. Full rehabilitation is delayed due to lack of funds and technical support.	(Skardu, Baltistan) – BAL–FARM–D2–02
What role do government departments play in	Government departments have occasionally supported lining and rehabilitation, but interventions are selective and limited by budget constraints. Cement lining is avoided in high-	(Skardu, Baltistan) – BAL–GOV–D2–03

maintenance and rehabilitation?	altitude areas due to cracking and short lifespan; earthen or hybrid solutions are preferred.	

C.2.3 Domain 3 – Water Allocation, Equity and Access

Question Asked	Verbatim Response	Reference
How is irrigation water distributed among households?	Water allocation is directly linked to landholding size. Larger landowners receive longer irrigation turns, while smaller landholders receive proportionally shorter durations. Despite this, the system is widely accepted as legitimate because it is transparent and historically rooted.	(Khaplu, Baltistan) – BAL–FARM–D3–01
How is water managed during scarcity periods?	During periods of adequate flow, water runs continuously and formal scheduling becomes unnecessary. Scarcity triggers stricter rotation, overseen by the water committee. Priority is given to staple crops and household food needs.	(Shigar, Baltistan) – BAL–FARM–D3–02
How is drinking water treated within the irrigation system?	Drinking water is not commodified within the irrigation system. Even households with little or no land are entitled to sufficient water for domestic use, reinforcing social cohesion and collective responsibility.	(Kharmang, Baltistan) – BAL–CBI–D3–03

C.2.4 Domain 4 – Climate Stress, Risk and Environmental Change

Question Asked	Verbatim Response (Expanded Description)	Reference
Has your irrigation system been affected by floods, landslides, or climate events?	Communities report increasing frequency and intensity of floods, which damage intakes, wash away channels, and delay irrigation during critical cropping periods. In some years, channels remain non-functional for several weeks, leading to crop failure.	(Skardu, Baltistan) – BAL–FARM–D4–01
What are the main sources of water and how have they changed?	Snowmelt remains the primary water source, with peak availability in July and August. However, variability has increased, making water supply less predictable. Early or late melting disrupts traditional agricultural calendars.	(Khaplu, Baltistan) – BAL–FARM–D4–02
What measures have been taken to reduce climate-related risks?	Preventive measures are minimal. Tree planting and informal embankments are used in some locations, but communities acknowledge that they lack the resources and technical knowledge to implement durable climate-resilient solutions.	(Kharmang, Baltistan) – BAL–CBI–D4–03

C.2.5 Domain 5 – External Support, Innovation and Future Needs

Question Asked	Verbatim Response (Expanded Description)	Reference
What external support have you	External support from NGOs and development programmes has been sporadic. Where provided, it has helped temporarily restore flows but has not	(Skardu, Baltistan) –

received for irrigation?	addressed long-term resilience or maintenance financing.	BAL–NGO–D5–01
Have you adopted any new technologies or coordination tools?	Use of mobile phones and WhatsApp groups has improved coordination for maintenance schedules and cash collection, especially involving non-resident landowners. However, no digital monitoring of flows or infrastructure exists.	(Shigar, Baltistan) – BAL–FARM–D5–02
What support would most strengthen your irrigation system?	Communities strongly expressed the need for government-supported rehabilitation of intakes, technical guidance adapted to mountain conditions, and an emergency fund mechanism to respond quickly after floods.	(Khaplu, Baltistan) – BAL–FARM–D5–03

Annex C.3: Summary of Primary Evidence: Diamer / Astore Region

C.3.1 Domain 1 – Institutional and Governance Arrangements

Question Asked	Verbatim Response (Expanded Description)	Reference
How are water rights allocated and exercised in your community?	Water rights are exercised according to land ownership under a long-standing customary system inherited from ancestors. Irrigation turns are proportional to the size of landholding, while drinking water is shared equally among all households. This system is accepted as legitimate and has been practiced for generations without written rules.	(Mani Bala / Pine Goner Farm, Diamer) – DIA–FARM–D1–01
Who supervises irrigation and manages water distribution?	A local water committee supervises water distribution. Previously a general village committee handled irrigation, but now a dedicated water resources committee oversees scheduling, monitoring, and coordination. Authority is based on community trust rather than formal appointment.	(Mani Bala, Diamer) – DIA–CBI–D1–02
How do non-resident or absentee landowners contribute?	Non-resident landowners are required to contribute financially because they cannot participate in physical labour. Contributions are calculated using local daily wage rates. This obligation is socially enforced and widely recognised as fair.	(Pine Goner Farm, Diamer) – DIA–FARM–D1–03

C.3.2 Domain 2 – Infrastructure Quality, Operation and Maintenance

Question Asked	Verbatim Response (Expanded Description)	Reference
Who decides on canal cleaning, repair, and labour arrangements?	It is already agreed within the community that one person from each household must participate in canal repair and cleaning whenever damage occurs, regardless of whether the water is used for irrigation or only for drinking. Households unable to send labour must pay cash instead.	(Mani Bala, Diamer) – DIA–FARM–D2–01
What are the most common technical problems affecting the irrigation channel?	The channel passes through sandy and unstable terrain, causing significant seepage losses. Water sinks into the ground, and temporary solutions such as filling gaps with grass are used. Because the channel is long, full control of leakage is not possible with local resources.	(Pine Goner Farm, Diamer) – DIA–FARM–D2–02
What role do government departments play in maintenance and rehabilitation?	The Water Management Department assesses damaged channels, carries out lining or earthen rehabilitation, and works jointly with communities. Cement lining is avoided at high altitudes due to rapid failure; earthen channels are preferred except where leakage is severe. Budget limitations constrain coverage.	(Water Management Dept., Diamer) – DIA–GOV–D2–03

C.3.3 Domain 3 – Water Allocation, Equity and Access

Question Asked	Verbatim Response (Expanded Description)	Reference
How is irrigation water distributed among households?	Water distribution follows land ownership. Households with larger landholdings receive longer irrigation turns, while smaller landowners receive proportionally shorter durations. Drinking water remains equally shared regardless of land size or family status.	(Mani Bala, Diamer) – DIA–FARM–D3–01
Is the available water sufficient for irrigation needs?	Water availability is seasonally sufficient, but channel size and damage limit effective delivery. During peak agricultural months, particularly July and August, insufficient channel capacity prevents water from reaching all fields adequately.	(Pine Goner Farm, Diamer) – DIA–FARM–D3–02
Is water distribution perceived as fair?	The distribution system is perceived as fair because it follows long-accepted rules. Inequality arises mainly due to infrastructure limitations rather than governance failures.	(Jawaharlal Nehruwadi, Diamer) – DIA–CBI–D3–03

C.3.4 Domain 4 – Climate Stress, Risk and Environmental Change

Question Asked	Verbatim Response (Expanded Description)	Reference
Has your irrigation system been affected by floods, landslides, or climate events?	Floods occur almost every year, especially in June and July, causing repeated damage. In the most recent event, the channel remained closed for 20–22 days. Crops failed, grass dried early, and households depended on government water tankers for drinking water.	(Mani Bala / Pine Goner Farm, Diamer) – DIA–FARM–D4–01
What are the main sources of irrigation water?	Snowmelt is the primary source of irrigation water. Water availability peaks in May, July, and August. If properly captured, this water could support crops during critical dry periods, but damaged intakes prevent effective use.	(Pine Goner Farm, Diamer) – DIA–FARM–D4–02
What preventive or adaptive measures exist?	There are no formal preventive measures. Communities hold meetings, encourage tree planting, and attempt minor protective works, but lack resources, advance funding, and technical designs to protect channels against floods.	(Diamer District) – DIA–CBI–D4–03

C.3.5 Domain 5 – External Support, Innovation and Future Needs

Question Asked	Verbatim Response (Expanded Description)	Reference
What external support has been received for irrigation?	External support has been limited. Some assistance was received from ETI and GBRSP during recent damage, but support is intermittent and insufficient to ensure long-term system resilience.	(Diamer District) – DIA–NGO–D5–01
Have any new technologies or coordination tools been adopted?	WhatsApp is used for coordination and communication. However, there has been little technological change in irrigation management, and no use of digital monitoring, mobile payments, or automation for water control.	(Mani Bala, Diamer) – DIA–FARM–D5–02

What support would most strengthen the irrigation system?	Communities emphasised the need for proper channel construction, intake protection, advance emergency funding for flood response, and closer coordination between government, community, and potential private partners.	(Diamer District) – DIA–FARM–D5–03

Annex D: Summary of Stakeholder Consultations and Key Inputs

Purpose of Annex

This annex presents a consolidated summary of stakeholder consultations conducted as part of the Irrigation Sector Study under the European Union–supported Policy Support Technical Assistance to the Government of Gilgit-Baltistan. It synthesises key inputs from primary field consultations across regions and stakeholder groups, highlighting dominant patterns, areas of convergence and divergence, and policy-relevant insights. Detailed, coded responses and verbatim evidence are presented separately in Annex C.

Overview of Stakeholder Consultations

Primary evidence for the study was generated through structured stakeholder consultations conducted across the regions of Gilgit, Baltistan, and Diamer/Astore. Consultations were designed to capture perspectives across the irrigation system lifecycle, including infrastructure development, operation and maintenance, governance arrangements, financing practices, equity considerations, and climate-related risks.

Stakeholders consulted included:

- Farmers and irrigation water users
- Customary water management actors (elders, mirabs, water committees)
- Community representatives involved in collective decision-making
- Local Government representatives and relevant line departments
- Institutions engaged in irrigation support, rehabilitation, or disaster response

Consultations were guided by structured instruments (see Annex A) and conducted through key informant interviews and group discussions. Responses were recorded, expanded, and coded to enable systematic synthesis and traceability.

Regional Context and Consultation Coverage

Gilgit Region

Consultations in the Gilgit region focused on relatively stable valley systems with long-established irrigation networks, but increasing exposure to climate-induced damage and infrastructure ageing. Stakeholders highlighted both the resilience of customary institutions and the growing limits of labour-based maintenance in larger command areas.

Baltistan Region

In Baltistan, consultations reflected strong collective traditions and high levels of community cohesion in irrigation management. Stakeholders emphasised the effectiveness of customary governance but noted rising pressures from extreme weather events, sedimentation, and the increasing technical complexity of repairs.

Diamer/Astore Region

Consultations in Diamer and Astore highlighted more fragile upper catchment systems, higher exposure to floods and landslides, and weaker institutional continuity due to migration and dispersed settlements. Stakeholders reported greater reliance on external assistance following major damage events.

Key Thematic Findings from Stakeholder Consultations

Institutional Arrangements and Governance

Across all regions, stakeholders consistently reported that customary institutions are the primary actors responsible for irrigation management. These institutions oversee water allocation, maintenance scheduling, labour mobilisation, and conflict resolution. Rules are largely unwritten but widely recognised and enforced through social norms.

Stakeholders emphasised that customary institutions derive legitimacy from community ownership and long-standing practice rather than formal legal recognition. While this has enabled effective local governance, it also limits access to public support, technical assistance, and financing.

Operation and Maintenance Practices

Routine operation and maintenance is overwhelmingly carried out through unpaid collective labour. Maintenance activities are typically seasonal or reactive, triggered by visible damage or imminent irrigation needs. Preventive maintenance is limited due to labour constraints and lack of technical support.

Stakeholders across regions noted that while collective labour remains functional, it is increasingly under strain due to migration, livelihood diversification, and ageing populations. Larger or more technically complex repairs are often beyond local capacity, leading to dependence on external assistance.

Financing and Cost-Sharing

Consultations revealed that monetary financing for routine O&M is rare. Cash contributions are typically mobilised only in exceptional circumstances, such as post-flood repairs or partial reconstruction. Even then, contributions are negotiated on a case-by-case basis and depend heavily on trust in local leadership.

In areas with prior engagement by development programmes, stakeholders reported limited experimentation with cost-sharing arrangements combining labour and cash. However, these practices have not evolved into recurrent financing mechanisms and often dissipate once external support ends.

Public Sector Engagement

Stakeholders consistently described public sector involvement in irrigation as episodic and project-driven. Government engagement is most visible during capital investment phases, rehabilitation works, or emergency response following disasters. Routine support for operation and maintenance is largely absent.

Local Government representatives acknowledged gaps in mandates, resources, and coordination mechanisms, particularly in relation to technical backstopping and post-disaster recovery. Stakeholders expressed a desire for clearer stewardship roles rather than direct operational control by public agencies.

Infrastructure Condition and Climate Risks

Stakeholders across all regions reported increasing vulnerability of irrigation infrastructure due to ageing channels, deferred maintenance, and heightened exposure to climate-related hazards. Floods, landslides, and glacial lake outburst floods were frequently cited as causes of major system damage.

Communities continue to rely on local knowledge and collective action to respond to shocks, but stakeholders emphasised that the scale and frequency of damage is exceeding local adaptive capacity. Climate risks are now viewed as a systemic challenge rather than episodic events.

Technology Adoption and Innovation

Consultations indicated limited adoption of modern irrigation technologies, including solar pumps, drip irrigation, and sprinkler systems. Where used, these technologies are applied in supplementary and context-specific ways, often disconnected from main irrigation systems.

Stakeholders noted that individual investment in efficiency technologies yields limited private benefit in collective systems, as water savings are redistributed downstream. This weakens incentives for adoption unless technologies are integrated at system level or linked to storage and land management objectives.

Equity, Inclusion, and Social Dynamics

Stakeholders reported that customary irrigation systems generally ensure equitable access to water within communities, though participation in decision-making is uneven. Women and youth are underrepresented in formal roles, despite playing important roles in agriculture and household water management.

Demographic change, including youth migration, was identified as a growing challenge for institutional continuity and labour mobilisation. Stakeholders expressed interest in more inclusive governance arrangements but emphasised the need for gradual, culturally appropriate change.

Cross-Cutting Insights

Across regions and themes, stakeholder consultations revealed several cross-cutting insights:

- Irrigation sustainability is shaped more by institutions, incentives, and collective capacity than by technical design alone
- Customary systems are resilient but increasingly strained by climate risk and socio-economic change
- Public sector roles are most effective when framed as stewardship and support, not operation
- Abrupt formalisation or externally imposed models risk undermining existing institutions
- Incremental, learning-oriented approaches are more likely to succeed than standardised reforms

Linkage to Analysis and Recommendations

The insights summarised in this annex directly inform the analytical findings presented in Sections 5 and 6 of the main report and underpin the legal and policy reform agenda set out in Section 8. Detailed, coded evidence supporting these findings is presented in Annex C, ensuring transparency and traceability from stakeholder inputs to policy recommendations.

Annex E: M&E Framework for the Legal and Policy Reform Agenda

Purpose

This Monitoring and Evaluation (M&E) framework tracks progress in implementing the proposed legal and policy reforms, with emphasis on institutional recognition, stewardship, financing, and adaptive learning. Monitoring focuses on process and institutional change rather than infrastructure outputs.

Reform Area	Key Indicators	Means of Verification	Frequency	Lead Responsibility
Recognition of Customary Institutions (8.1)	Legal/regulatory provisions recognising customary water institutions; documented interfacing with public agencies	Gazette notifications; administrative records	Annual	LG&RD; Law Dept.; District Admin
Community Water Management Utilities (8.2)	Enabling legal provisions issued; pilot community utility entities operating	Legal texts; pilot programme reports	Annual	LG&RD; P&D; Local Governments
Public Sector Stewardship Roles (8.3)	Mandates clarified; stewardship or coordination guidelines in place	Policy notifications; departmental guidelines	Annual	LG&RD; Irrigation Dept.; P&D
Financing and Public Support (8.4)	Policy instruments enabling predictable support; targeted funding provided	Budget documents; programme reports	Annual	Finance Dept.; P&D; Local Governments
Social Safeguards and Inclusion (8.5)	Inclusion and transparency provisions embedded; women/youth participation	Legal texts; institutional records	Biennial	LG&RD; Social Welfare; Local Governments
Learning-Oriented Legal Adaptation (8.6)	Pilot initiatives under adaptive arrangements; policy revisions informed by pilots	Pilot evaluations; policy review notes	Every 2–3 years	Law Dept.; P&D; Policy Units

Monitoring and Learning Modality

Monitoring will be conducted annually to track legal and institutional milestones. Periodic evaluations will assess institutional performance and reform effectiveness. Learning from pilot initiatives will inform iterative refinement of legal and policy provisions.

Annex F: Alignment Matrix of Irrigation Sector KPIs with SDG Targets

Purpose of Annex

This annex presents an alignment matrix linking key irrigation sector performance indicators to relevant Sustainable Development Goal (SDG) targets. The matrix is intended to demonstrate how improvements in irrigation governance, operation and maintenance, resilience, and inclusion contribute to national and international development commitments. The alignment supports policy coherence, results-based monitoring, and EU reporting requirements under the Policy Support Technical Assistance.

Scope and Approach

The matrix focuses on system-level and institutional KPIs rather than purely technical efficiency metrics. Indicators are aligned with the realities of community-managed irrigation systems in Gilgit-Baltistan and are consistent with the study's emphasis on governance, sustainability, resilience, and equity. SDG targets are referenced at target level rather than indicator code level to maintain clarity and policy relevance.

Table F.1: Alignment Matrix of Irrigation Sector KPIs with SDG Targets

Irrigation Sector KPI	Description	Relevant SDG	SDG Target
Functioning community-based O&M institutions	Share of irrigation systems with active customary or community water management institutions	SDG 6	6.5 – Implement integrated water resources management at all levels
Reliability of irrigation service delivery	Continuity of water supply during irrigation season	SDG 2	2.4 – Ensure sustainable food production systems
Maintenance response time	Time taken to mobilise labour or resources following system damage	SDG 9	9.1 – Develop reliable and resilient infrastructure
Share of systems with documented governance arrangements	Presence of locally recognised rules or agreements for water allocation and O&M	SDG 16	16.6 – Develop effective, accountable, and transparent institutions
Public technical backstopping availability	Access to technical support from public institutions when required	SDG 6	6.a – Expand international cooperation and capacity-building for water-related activities
Climate risk preparedness of irrigation systems	Presence of preventive or adaptive measures addressing floods, landslides, and GLOFs	SDG 13	13.1 – Strengthen resilience and adaptive capacity to climate-related hazards
Equity in water access	Perceived fairness in water distribution across users and locations	SDG 10	10.2 – Promote social, economic, and political inclusion

Participation of women and youth in irrigation governance	Representation of women and youth in decision-making roles	SDG 5	5.5 – Ensure full participation and equal opportunities for leadership
Financial sustainability of O&M	Ability to mobilise labour and/or modest financial resources for maintenance	SDG 1	1.4 – Ensure equal access to basic services
Adoption of context-appropriate irrigation innovations	Use of technologies aligned with local institutional and environmental conditions	SDG 12	12.2 – Achieve sustainable management and efficient use of natural resources

Interpretation and Use

The alignment matrix demonstrates that irrigation sector performance in Gilgit-Baltistan contributes directly to multiple SDGs, particularly those related to water management, food security, climate resilience, institutional effectiveness, and social inclusion. The emphasis on institutional and governance KPIs reflects the study's finding that sustainable irrigation outcomes depend primarily on collective capacity and stewardship rather than technical design alone.

The matrix is intended for use as:

- A policy coherence tool linking sector reforms to SDG commitments
- A monitoring reference for government and development partners
- A reporting aid for EU-supported policy dialogue and results frameworks

Limitations

The KPIs presented are qualitative or semi-quantitative in nature and are not intended to replace national SDG indicator reporting. They are designed to support sector-level monitoring and learning within the scope of the Policy Support Technical Assistance.

Bibliography

This bibliography brings together the principal policy documents, legislation, programme reports, and academic literature that informed the analysis presented in this study. The references reflect a combination of global and regional scholarship on irrigation governance, collective action, and institutional reform, alongside country- and region-specific sources relevant to Gilgit-Baltistan and Pakistan. The literature has been used to contextualise and interpret primary field evidence, support comparative analysis, and inform the legal and policy reform agenda proposed under this Technical Assistance.

References

Abdullaev, I., Kazbekov, J., Manthritilake, H., & Jumaboev, K. (2010). Water user groups in Central Asia: Emerging forms of collective action in irrigation. Water Resources Management, 24(5).

Bilal, A., & Khan, T. (2003). Customary water management and indigenous institutions in Northern Pakistan. IUCN Pakistan.

Boelens, R. (2015). Water, power and identity: The cultural politics of water in the Andes. Routledge.

FAO. (2017). Modernizing irrigation management: The role of institutions and governance. FAO Water Reports. Food and Agriculture Organization of the United Nations.

Government of Gilgit-Baltistan. (2024). Gilgit-Baltistan Water Users' Associations Act, 2024.

Government of Gilgit-Baltistan. (2024). Gilgit-Baltistan River Protection Act, 2024.

Government of Pakistan. (2018). National Water Policy. Ministry of Water Resources.

ICIMOD. (Various years). Climate and water resources studies in the Hindu Kush–Himalaya region. International Centre for Integrated Mountain Development.

IFAD. (2016). Sustainable smallholder irrigation development: Lessons from project experience. International Fund for Agricultural Development.

IUCN Pakistan. (2000). Customary law and natural resource management in the Northern Areas of Pakistan. IUCN Pakistan Country Office, Islamabad.

IUCN Pakistan. (2003). Natural resource governance and community institutions in Northern Pakistan. IUCN Pakistan.

Khan, T., & Hunzai, I. (2000). Sharing water: Irrigation and water management in the Karakoram. [Publisher not specified].

Kreutzmann, H. (1996). Pamirian crossroads: Kirghiz and Wakhi of High Asia. Oxford University Press.

Kreutzmann, H. (2011). Pastoral practices and irrigation in the Karakoram. In Pastoralism and mountain livelihoods in Central Asia (pp. xx–xx). [Publisher not specified].

Ostrom, E. (1992). Crafting institutions for self-governing irrigation systems. Institute for Contemporary Studies Press.

PCRWR. (2023). National Water Conservation Strategy 2023–27. Pakistan Council of Research in Water Resources.

Pradhan, P. (2010). Patterns of irrigation organization in Nepal: A comparative study of farmer-managed and agency-managed systems. International Water Management Institute.

Sehring, J. (2009). The politics of water institutional reform in neopatrimonial states: A comparative analysis of Kyrgyzstan and Tajikistan. VS Verlag für Sozialwissenschaften.

World Bank. (2015). Sustaining irrigation services: Institutional and financing reforms. World Bank Water Global Practice.